

Data catalog tools

Citation for published version (APA):

Tonnarelli, M., Kumara, I., Driessen, S., Tamburri, D. A., van den Heuvel, W. J., & Oor, P. (2025). Data catalog tools: A systematic multivocal literature review. *Journal of Systems and Software*, 230, Article 112584. <https://doi.org/10.1016/j.jss.2025.112584>

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DOI:

[10.1016/j.jss.2025.112584](https://doi.org/10.1016/j.jss.2025.112584)

Document status and date:

Published: 01/12/2025

Document Version:

Publisher's PDF, also known as Version of Record (includes final page, issue and volume numbers)

Please check the document version of this publication:

- A submitted manuscript is the version of the article upon submission and before peer-review. There can be important differences between the submitted version and the official published version of record. People interested in the research are advised to contact the author for the final version of the publication, or visit the DOI to the publisher's website.
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Data catalog tools: A systematic multivocal literature review[☆]

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ARTICLE INFO

Keywords:

Data catalogs
Metadata management
Big data architectures
Data mesh
Data lake
Data lakehouse
Life-cycle management
Data governance

ABSTRACT

A data catalog enables an organization to maintain an inventory of its data assets by collecting and managing the relevant metadata. We conducted a systematic multi-vocal literature review on data catalogs to understand their features and usage. We systematically selected and analyzed 86 literature sources and 39 catalog tools. We first utilized the findings from the literature to develop a classification framework comprising 24 fine-grained and five high-level features, along with three maturity levels. Next, we analyzed 39 tools based on the classification framework. Organizations typically include a data catalog as a component in their big data platforms and use it to support the various phases of the metadata management lifecycle. Hence, we also mapped the catalog features to the requirements of metadata-driven big data architectures, namely data mesh, data lake, and data lakehouse. Moreover, the mappings of the features to the phases in a metadata management lifecycle were developed. Our findings shall aid organizations in making informed decisions when choosing data catalog tools and help researchers identify the critical research issues in data cataloging and metadata management.

Editor's note: Open Science material was validated by the Journal of Systems and Software Open Science Board.

1. Introduction

Metadata, often described as *data about data*, are critical in managing and understanding large and complex data assets (Furner, 2019; Sen, 2004). Metadata include information such as schemas, origin, history, location, ownership, permissions, and versions of data (Furner, 2019; Gartner, 0000). Such information enables discovery, understanding, and governance of data assets within and across organizations. A lack of metadata is a significant bottleneck to turning data into valuable assets in data-driven organizations (Graetsch et al., 2023).

Metadata are also pivotal in big data architectures like data mesh, data lake, and data lakehouse. In the data mesh, which is a decentralized approach where each data domain is responsible for its data products, metadata facilitates the interoperability and discoverability of data products across domains (Dehghani and Fowler, 2022; Goedegebuure et al., 2024). In the data lakehouse, which combines data lakes and data warehouse functionalities, metadata support efficient data storage, querying, and analytics (Zaharia et al., 2021). In the data lake architecture, which stores vast amounts of structured, semi-structured,

and unstructured data in its raw form, metadata play a crucial role in achieving data lineage, discovery, and governance (Jahnke and Otto, 2023a).

Data catalogs are emerging as an essential component in big data architectures to acquire, store, and manage metadata (Gröger, 2021). A data catalog is a repository of metadata that has been acquired from various data sources in an organization (Olesen-Bagneux, 2023). For example, data scientists in a manufacturing company need to forecast the demand for products for the next quarter. With a data catalog, they can quickly find relevant data sets like sales figures, inventory levels, and supply chain metrics through the company's data catalog by searching data sets based on their metadata. Without a catalog, they would spend significant time manually searching across departments for this information, risking the use of outdated or incomplete data. Data catalogs also promote collaboration and data sharing among teams, fostering a data-driven culture (Olesen-Bagneux, 2023) within organizations.

[☆] Editor: Heiko Koziolk.

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While many data catalog tools exist, such as Alation,¹ Atlan,² Informatica Data Marketplace,³ DataHub⁴ and CKAN,⁵ there is a lack of systematic study focusing on their features, usage in metadata management, and integration with data architectures such as data mesh, data lake, and data lakehouse. Organizations can use such information to select a suitable data catalog tool and successfully implement their metadata management use cases. Researchers can use the results from this study to understand the gaps between industry and academia and identify the features of data catalogs that require further investigation.

In this research, we aim to study data catalog tools to understand their characteristics and formulate guidelines for using them to manage metadata in big data architectures. For this purpose, we conducted a Systematic Multivocal Literature Review (MLR) (Garousi et al., 2019) on data catalog and metadata management tools. An MLR enables us to conduct a wide range of research across various resources, allowing us to cover the latest academic results and the most recent industry findings.

We analyzed 86 sources, from white and grey literature, along with 39 data catalogs available on the market and their respective documentation. We identified 24 fine-grained features and 5 high-level ones for data catalogs, and we mapped them to the metadata lifecycle management steps, 3 maturity levels for data catalogs, and the requirements of modern metadata-driven big-data architectures, such as data mesh.

The paper is organized as follows. Section 2 provides an overview of existing research on data catalog and metadata management tools. Section 3 defines the research questions and describes the methodology adopted. In Section 4, we describe the classification framework we derived from our MLR, and we identify the features of data catalogs. In Section 5, we classify data catalogs available on the market according to the classification framework. Section 6 maps the features to metadata life-cycle management steps, and Section 7 maps the features to big data architecture requirements. Section 8 discusses our results. Finally, we describe the threats to the validity of our study in Section 9 and draw our conclusions in Section 10.

2. Background and related work

In this section, we introduce the key types of metadata, including business, technical, and operational, as they are fundamental concepts underlying data catalogs. We also review related work on data marketplaces, integration with modern data architectures, essential functionalities of data catalog tools, and the role of data catalogs in managing metadata. The section concludes by identifying research gaps and motivating the need for a comprehensive analysis of data catalog capabilities and their support for the metadata lifecycle.

2.1. Metadata

The metadata of an artifact (e.g., a data point, data set, or machine learning model) provide information about one or more aspects of the artifact. A data catalog ingests and manages various types of metadata from different data sources, including structured, semi-structured, and unstructured data. While many metadata classifications exist (Baca, 2008; Zeng and Qin, 2020; Sawadogo and Darmon, 2021a; Ravat and Zhao, 2019), which include *administrative*, *descriptive*, *preservation*, *technical* and *use* metadata, for brevity we will describe them according to the Data Management Body of Knowledge (DAMA-DMBOK) (DAMA International, 2017), which identifies three primary categories: business metadata, technical metadata, and operational metadata.

Business metadata. Contextual and semantic meaning to data is provided by business metadata, making them interpretable for business users. They include business glossaries, data ownership information, regulatory compliance labels, and data quality rules. This type of metadata helps ensure that data is aligned with business processes and regulatory requirements within organizations. For example, a Customer ID field in a customer relationship management (CRM) system may be defined as a unique identifier for each customer in the business glossary. Additionally, it may be classified as personally identifiable information (PII) and governed by the General Data Protection Regulation (GDPR) policies, restricting access to authorized personnel only. Business metadata also include data classifications (e.g., “Confidential,” “Public,” “Restricted”), which are essential for security and compliance purposes.

Technical metadata. Technical metadata describe the structural aspects of data and thus helps data engineers and data scientists to correctly store, manage, and use data. They encompass schema definitions, data types, constraints, indexing strategies, and storage details. These metadata enable systems to understand how data is stored, processed, and integrated across platforms. For instance, in a cloud data lake, a dataset might be stored in a storage bucket as partitioned Parquet⁶ files, with schema details specifying column names, data types, and constraints such as primary and foreign keys.

Operational metadata. Operational metadata provide insights into processing aspects of data at run time. They include query execution logs, data freshness indicators, access control logs, and data lineage. This type of metadata is critical for monitoring data quality, optimizing system performance, and ensuring compliance with security policies. For example, a dataset used in a financial reporting dashboard may contain metadata on last update timestamps, query response times, and user access logs, helping organizations ensure that reports are based on the most recent and accurate data.

2.2. Related work

Several studies have attempted to review and analyze the tools used in data cataloging and metadata management.

2.2.1. Data marketplaces, integration with data architectures, and metadata-driven technologies

A concept related to the data catalog is the data marketplace, a platform that provides the infrastructure and services for exchanging data products between providers and consumers across diverse environments, incorporating features for dataset organization, access, sharing, and pricing (Fernandez et al., 2020; Niu et al., 2018; Barns, 2018). Driessen et al. (2022) systematically review data market platforms, identifying key application domains where data markets are actively used and introducing taxonomies of design challenges and proposed solutions. Palviainen and Suksi (2023) conduct a large-scale survey of data marketplaces, analyzing over 190 commercial platforms. They identify key metadata challenges in data trading ecosystems, particularly the lack of interoperability standards for metadata structuring and pricing mechanisms. Sharma et al. (2020) propose a blockchain-based data marketplace architecture, leveraging smart contracts and decentralized data storage to enable secure metadata exchange.

Zhang et al. (2023) focus on data pricing models, demonstrating how metadata influence pricing in data exchange platforms. The role of data marketplaces in metadata lifecycle management is becoming increasingly relevant. The study on trusted data marketplaces for credit scoring (Azcoitia and Laoutaris, 2022) outlines a reference architecture that integrates data catalogs with privacy-enhancing technologies, such as homomorphic encryption and multi-party computation. Similarly,

¹ <https://www.alation.com>

² <https://atlan.com/p/data-catalog-atlan/>

³ <https://www.informatica.com/nl/products/data-governance/cloud-data-marketplace.html>

⁴ <https://datahubproject.io>

⁵ <https://ckan.org>

⁶ <https://parquet.apache.org/>

the review of data markets in smart cities and edge computing (Kučera et al., 2013) explores how metadata governance enables real-time, decentralized data exchange in urban environments.

Data catalogs do not operate in isolation: they must be integrated with enterprise data architectures and frameworks to support efficient data discovery and exchange in organizations. Recent research emphasizes the pivotal role of data catalogs across data architectures such as data lakes, data lakehouses, and data mesh. For example, Nargesian et al. (2019) argue that without metadata management systems, data lakes risk becoming unmanageable *data swamps*. They show how metadata can improve data discovery and reuse in data lakes (using catalogs). In the context of time-series data scenarios, Pohl et al. (2024) identify the metadata management layer as a key component in the data lakehouse architecture to support data governance and transaction management. Nambiar and Mundra (2022) discuss the roles of metadata in both data warehouses and data lakes. They advocate for *knowledge catalogs* that enhance semantic understanding, data lineage, and governance across structured and unstructured data environments. Karkošková (2024) considers the data catalog to be the backbone of the data mesh, enabling decentralized governance and cross-domain collaboration through features such as data shops, knowledge graphs, and lineage tracking. Other studies also suggest using the data catalog to support data product discovery in data mesh (Van Eijk et al., 2024; Goedegebuure et al., 2024).

2.2.2. Functionalities and requirements of data catalogs

A fundamental challenge in data cataloging is identifying the key functionalities required for effective metadata management. Jahnke and Otto (2023b) provide an in-depth overview of modern data catalog architectures, outlining essential features for metadata management, data discovery, and governance. However, they do not offer a general framework for classifying tools, leaving room for further exploration. Petrik et al. (2023) take a more structured approach by identifying 13 functional requirements for enterprise data catalogs, categorizing them into metadata management, data governance, interoperability, and intelligent automation. Beyond catalog functionalities, metadata interoperability and automation are crucial in ensuring efficient data exchange. Labadie et al. (2020) discuss how data catalogs can operationalize the FAIR (Findable, Accessible, Interoperable, Reusable) principles, particularly within enterprise settings. Similarly, Ehrlinger et al. (2021) present 4 functional components of data catalogs and an 8-step guideline for their implementation.

Data catalogs could also support complex metadata governance and lifecycle management. Sawadogo and Darmont (2021a) provide an overview of the state-of-the-art regarding different approaches for data lake design, focusing on architectures and metadata management. Highlighting the importance of metadata management in data lakes, Megdiche et al. (2021) propose and validate a metadata model compliant with the FAIR (Findable, Accessible, Interoperable, and Reusable) principles. The study by Thamjaroenporn et al. on big data analytics in digital government introduces a framework for data governance and cataloging, emphasizing data exchange and metadata standardization (Thamjaroenporn and Achalakul, 2020). Martinez-Gil's work on quantifying Open Data Catalog (ODC) similarity explores methods for comparing metadata structures and ensuring interoperability across different catalogs (Martinez-Gil, 2023).

2.3. Research gaps and goals

Despite the growing importance of data catalogs and metadata management, several key issues remain unresolved. First, there is a limited understanding of the core features of data catalogs and the tools available to support them. Additionally, the integration of data catalogs with metadata-centric architectures, such as data mesh, data lake, and data lakehouse, is not well-defined. Another gap is a systematic analysis of the extent to which current data catalog tools support the entire

metadata life cycle. Existing studies tend to focus on individual aspects of metadata or catalogs without providing a comprehensive view that addresses all these elements together.

We aim to bridge these research gaps by systematically reviewing data catalog tools using both academic and grey literature sources. This study analyzes the core features of data catalogs, reviews existing tools, and explores their alignment with metadata-driven architectures. We also evaluate how data catalogs support the different phases in the metadata lifecycle, providing a comprehensive perspective on data cataloging to guide future research and practical applications by practitioners.

3. Research methodology

We conducted a Systematic Multivocal Literature Review (MLR) (Garousi et al., 2019) to identify and classify a list of data catalog tools and their features and capabilities. We chose MLR because most data catalog tools are from the industry, and many related grey literature sources exist. Our MLR primarily follows the guidelines proposed by Garousi et al. (2019) and MLR studies (Islam et al., 2019a; Recupito et al., 2022; Islam et al., 2019b; Edison et al., 2021; Islam et al., 2019b). We also drew inspiration and guidance from literature reviews and comparative studies of tools (Giamattei et al., 2024; Yussupov et al., 2021).

Fig. 1 shows our MLR methodology. We followed a structured process, starting with a pilot study to assess feasibility and define an MLR approach covering both white and grey literature. Four research questions guided the study, focusing on data catalog capabilities, tools, metadata support, and relevance to modern architectures. We applied inclusion/exclusion criteria and designed two queries to retrieve sources, refining them through filters and time constraints. This yielded 86 sources: 54 academic papers, 23 grey literature sources, and nine sources identified through snowballing. Of the 65 identified tools, 39 were selected after the filtering process. A classification framework was built from extracted features and mapped to metadata lifecycle stages and big data architecture needs.

3.1. Pilot study and planning: Background, research questions and literature selection criteria

This paper aims to investigate the state of the art of data catalog tools, assess their main capabilities and features, and guide organizations in choosing the most suitable tool for building a data catalog. We also consider the use of such tools within emerging data architectures, such as data lakehouses, data meshes, and data lakes.

The pilot study was the first step, used to test methods and define the scope of our research. We reviewed the studies that analyze the academic literature on data catalogs and metadata management, and found limited coverage of the dimensions we aimed to explore. As a result, we included the relevant grey literature, leading to an MLR approach.

We consider the following four research questions in our MLR:

1. **RQ₁.** *What are the key capabilities of data catalogs?* This question aims to identify and synthesize key features provided by data catalogs, providing a taxonomy for systematically classifying and comparing the cataloging tools.
2. **RQ₂.** *What tools can be used to build a data catalog?* We want to provide an overview of the current state of the art regarding tools that enable enterprises to build data catalogs. We also aim to analyze the selected tools based on the features identified in RQ₁.
3. **RQ₃.** *To what extent can the data catalog tools support the phases in the metadata management life-cycle?* We want to understand how the various features of cataloging tools can be employed to support various metadata management activities.

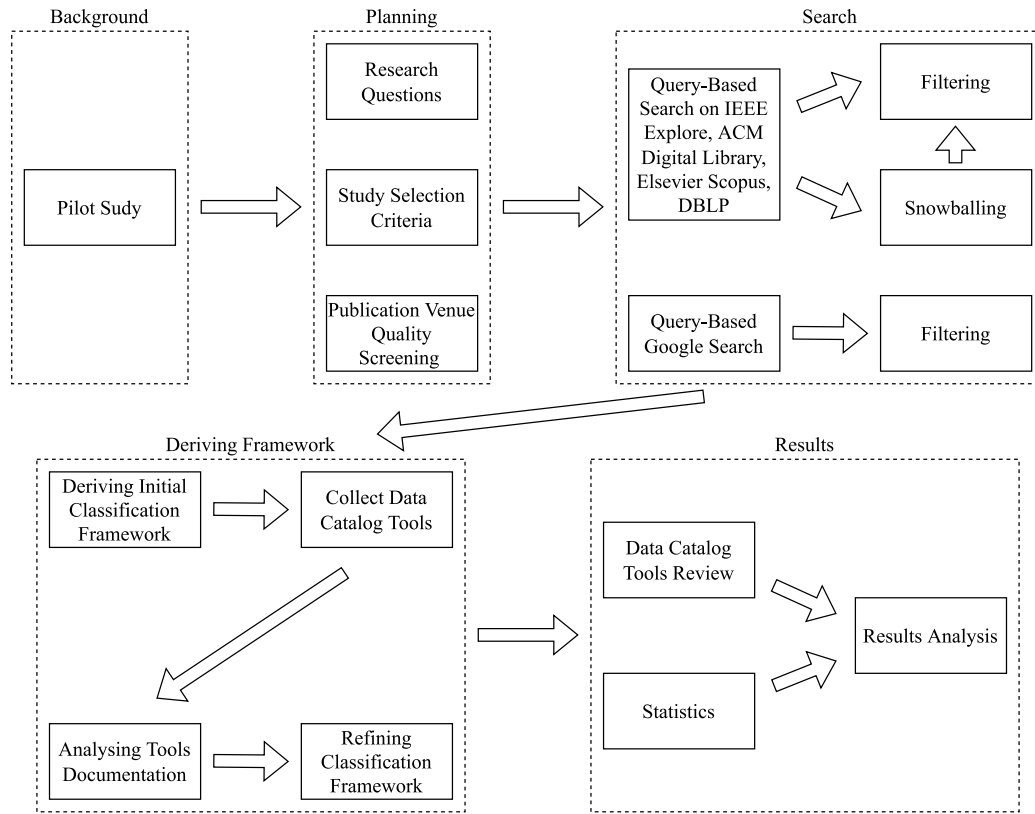


Fig. 1. Multivocal literature review methodology.

4. **RQ₄.** *How can the data catalog tools be employed within big data architectures?* This question aims to understand how data catalog tools can be used to implement the metadata requirements of modern metadata-driven big data architectures.

After defining the research questions, we established a set of inclusion and exclusion criteria to filter out the results of our research queries and tools, which are summarized in Table 1.

3.2. Search: Queries, filtering and snowballing

We defined two search queries to collect literature addressing our research questions. The search strings we used are structured as follows:

$$\begin{aligned}
 & (data \wedge catalog) \vee \\
 & (metadata \wedge (tool \vee framework \vee \\
 & platform \vee repository \vee store \vee registry)) \\
 & \wedge PUBYEAR > 2014 \wedge PUBYEAR < 2025 \\
 & data \wedge catalog \wedge tool
 \end{aligned}
 \tag{1}$$

Several pilot searches helped refine our terms and ensure we captured relevant studies. Query (1) was designed for academic digital libraries and built using terminology from prior studies on data catalogs (Jahnke and Otto, 2023b; Petrik et al., 2023; Ehrlinger et al., 2021), along with related and synonymous concepts. Terms were combined using AND/OR logic and applied to article titles, abstracts, and keywords. We followed the methodology of several recent MLRs (Xiong et al., 2023; Islam et al., 2019a; Recupito et al., 2022), and selected four major academic libraries: ACM Digital Library,⁷ IEEE Xplore,⁸ Elsevier

Scopus,⁹ and DBLP.¹⁰ These platforms were chosen as they collectively index key journals and conferences in data and metadata management.

Initially, we considered a 5-year publication window but extended it to 10 years after identifying older, foundational studies useful for building our classification framework and taxonomy.

The initial execution of Query (1) yielded an overly large result set (e.g., over 9000 entries from ACM DL alone), so we reduced the scope using venue-based filtering, inspired by the approach of Xiong et al. (2023). We included 17 journals and 27 conferences, all ranked A or A* in the CORE Conference Portal,¹¹ and Journals from the CORE Journal Portal,¹² as shown in Tables 2 and 3. This filtering reduced the total number of results across all libraries to around 2100. We focused on sources from fields such as software engineering, data engineering, information science, and databases. Biomedical and Health Informatics was also included due to the presence of relevant studies in high-quality venues in that domain.

Query (2) targeted grey literature through the Google¹³ search engine. Given the rise in popularity of data catalogs, conventional metadata management tools are often rebranded under this label. Thus, a complex query was not necessary. We screened the top 100 results (excluding advertisements), in line with common practices, showing that the most relevant content appears within the first few pages (Islam et al., 2019b; Garousi and Mäntylä, 2016)

Fig. 2 details the number of results per platform. From grey literature, we retained 23 sources (mostly blogs and vendor websites) after filtering. From white literature, we initially identified 2013 papers and narrowed these down to 54 after two filtering steps: screening titles and

⁹ <https://www.scopus.com/home.uri>

¹⁰ <https://dblp.org/>

¹¹ <https://portal.core.edu.au/conf-ranks/>

¹² <https://portal.core.edu.au/jnl-ranks/>

¹³ <https://www.google.com/>

⁷ <https://dl.acm.org/>

⁸ <https://ieeexplore.ieee.org/Xplore/home.jsp>

Table 1
Inclusion and exclusion criteria.

Category	Criteria	Inclusion/ Exclusion	Motivation
White Literature	Discusses metadata management topics	Inclusion	To ensure coverage of various metadata management practices and functionalities.
White Literature/Grey Literature	Written in English	Inclusion	To maintain accessibility and readability for a wide academic audience.
White Literature/Grey Literature	Introduces novel practices or frameworks used with data catalogs	Inclusion	To advance the knowledge of cutting-edge methodologies in data cataloging.
Grey Literature	Vendor blogs/websites	Exclusion	To avoid potential bias and promotional content that may not be academically rigorous.
White Literature	Not covering metadata management tools/practices	Exclusion	To keep the focus relevant to the specific field of data cataloging.
White Literature	Sponsored Content	Exclusion	To eliminate bias introduced by sponsorship, which might affect the objectivity of the information.
Grey Literature	Results beyond the first 10 Google pages	Exclusion	To focus on more readily accessible and likely more relevant sources.
Grey Literature	Discusses one or more tools	Inclusion	To ensure that different tools and their specific applications are covered.
White Literature/Grey Literature	Source not accessible	Exclusion	To ensure all referenced materials are available for review and use by researchers.
White Literature	Papers shorter than three pages	Exclusion	To ensure substantial content and depth in the discussion.
White Literature	Lacks a defined research methodology	Exclusion	To maintain academic rigor and reliability of the research presented.
Tools	Documentation is in English	Inclusion	To facilitate understanding and use by the international research community.
Tools	Open source and closed source	Inclusion	To provide a comprehensive view of available tools regardless of their licensing.
Tools	Supports the collection of at least technical metadata	Inclusion	To ensure the tool meets basic functional requirements for data cataloging.
Tools	Declared as discontinued or unmaintained	Exclusion	To focus on tools that are current and supported.
Tools	Primary focus is not data cataloging	Exclusion	To maintain relevance to the specific area of data cataloging.

Table 2
Journals included in the search query scope.

Venue name	Search target			
	ACM	IEEE	DBLP	SCOPUS
ACM Transactions on Information Systems	✓			
Data Mining and Knowledge Discovery				✓
Information and Software Technology				✓
Information Processing and Management				✓
Information Sciences				✓
ACM Transactions on Database Systems	✓			
Annual Review of Information Science and Technology				✓
IEEE Journal of Biomedical and Health Informatics		✓		
IEEE Transactions on Information Theory		✓		
IEEE Transactions on Knowledge and Data Engineering		✓		
Information Systems				✓
Journal of Information Technology				✓
IEEE Transactions on Software Engineering		✓		
ACM Transactions on Software Engineering and Methodology	✓			
Journal of the Association for Information Science and Technology				✓
Journal of Systems and Software				✓
Empirical Software Engineering				✓

Table 3
Conferences included in the search query scope.

Venue name	Search target			
	ACM	IEEE	DBLP	SCOPUS
ACM International Conference on Information and Knowledge Management (CIKM)	✓			
ACM International Conference on Knowledge Discovery and Data Mining (KDD)	✓			
ACM International Conference on Research and Development in Information Retrieval (SIGIR)	✓			
ACM International Conference on the Foundations of Software Engineering (was ESEC/FSE, now FSE)	✓			
ACM SIGMOD-SIGACT-SIGART Conference on Principles of Database Systems (PODS)	✓			
ACM Special Interest Group on Management of Data Conference (SIGMOD)	✓			
Automated Software Engineering Conference (ASE)			✓	✓
Conference on Innovative Data Systems Research (CIDR)			✓	✓
European Conference on Information Retrieval (ECIR)			✓	✓
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Database			✓	✓
European Conference on Software Architecture (ECSA)			✓	✓
Extending Database Technology (EDBT)			✓	✓
Foundations of Software Science and Computational Structures (FOSSACS)			✓	✓
IEEE International Conference on Data Mining (ICDM)		✓		
IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)		✓		
IEEE International Conference on Software Maintenance and Evolution (prior to 2014 was ICSM, now ICSME)		✓		
IEEE International Working Conference on Mining Software Repositories (MSR)		✓		
International Conference on Advanced Information Systems Engineering (CaiSE)			✓	✓
International Conference on Data Engineering (ICDE)			✓	✓
International Conference on Database Theory (ICDT)			✓	✓
International Conference on Evaluation and Assessment in Software Engineering (EASE)			✓	✓
International Conference on Learning Analytics and Knowledge (LAK)			✓	✓
International Conference on Software Architecture (was WICSA, now ICASA)			✓	✓
International Conference on Software Engineering (ICSE)			✓	✓
International Conference on Very Large Databases (VLDB)			✓	✓
International Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS)			✓	✓
International Symposium on Software Reliability Engineering (ISSRE)			✓	✓

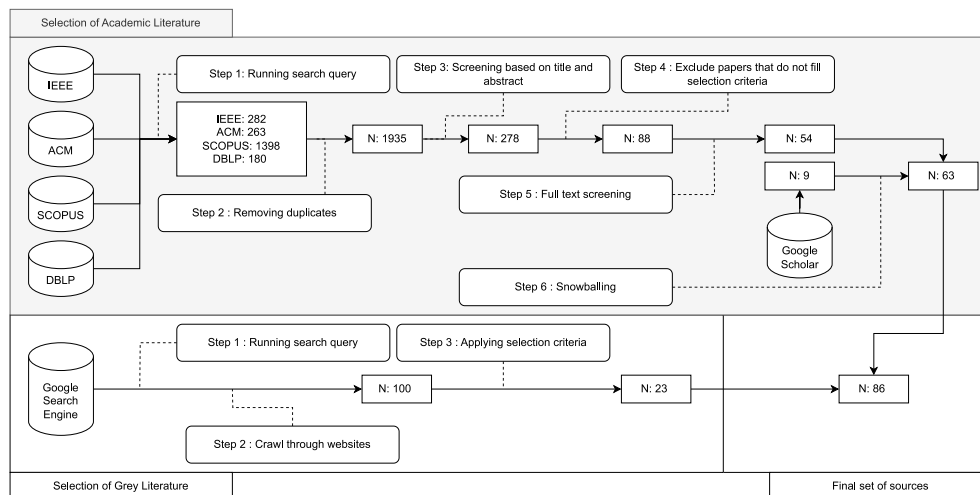


Fig. 2. Study selection process of our MLR.

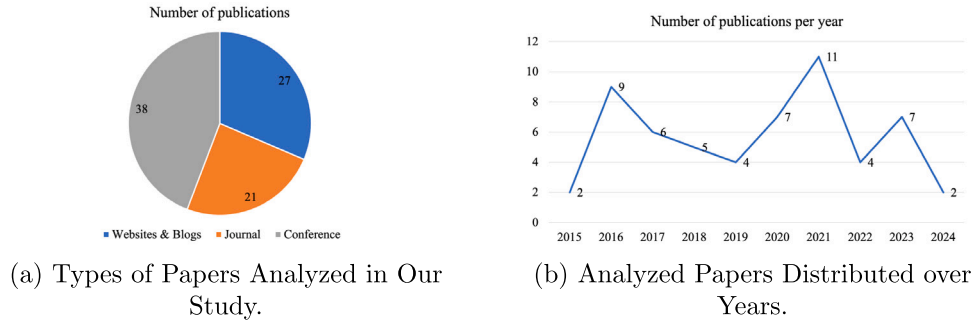


Fig. 3. Data catalog tools general information.

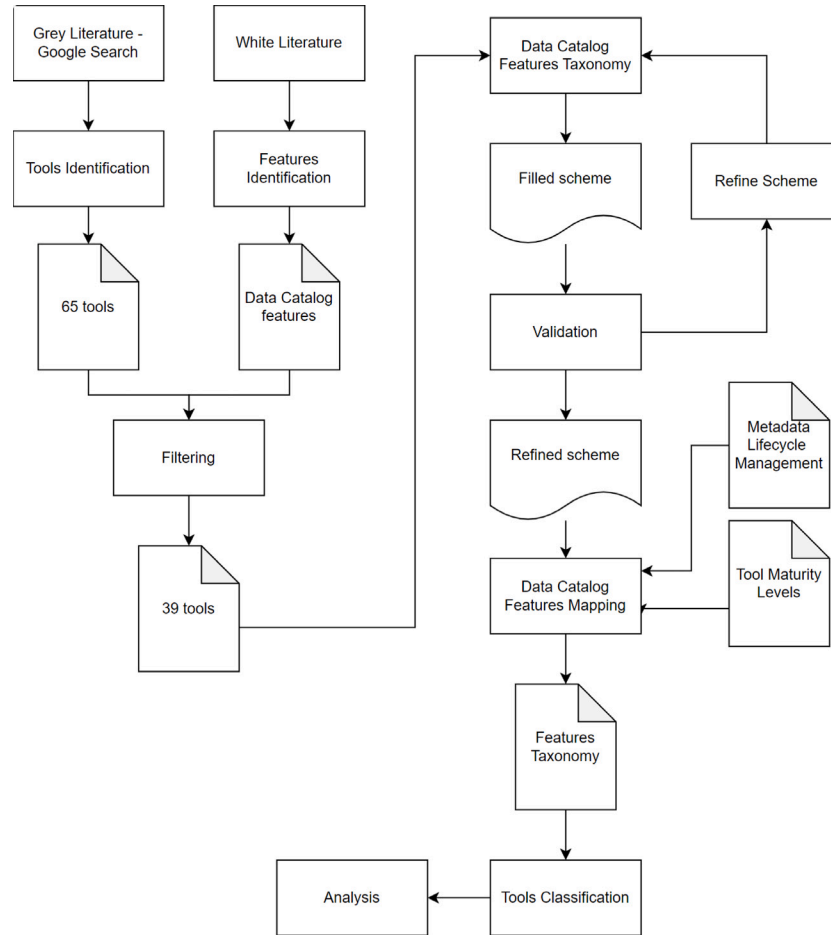


Fig. 4. Deriving classification framework process.

abstracts, followed by a full-text review. We also applied backward and forward snowballing, which yielded an additional 9 relevant papers. In total, we included 86 sources (54 white literature and 32 grey literature) in our study. Fig. 3 summarizes the publication venues and temporal distribution of the selected papers.

3.3. Deriving framework: Classification framework and tool analysis

In this section, we describe the process we followed to derive the classification framework, similar to the approach adopted by Giamattei et al. (2024). Fig. 4 depicts the process. The first steps proceeded in parallel: we extracted a list of tools from the grey literature results and identified an initial list of features from the white literature papers.

Initially, we identified a set of 65 tools from various sources (Lawton, 2024; Club, 2024; Anon, 2024a; King, 2024a; Anon, 2024b; Barjatiya, 2024; Anon, 2024c; Travkin, 2024; Reddit, 2024; King, 2024b; Beatrice, 2024a; K., 2024; J., 2024; Abdullahi, 2024; Mohmad, 2024a; Torcato, 2024; Beatrice, 2024b; Kumar, 2024; Mohmad, 2024b; research@theseattledataguy.com, 2024; Preethi, 2024; Skogstad, 2024). We used the features as filters to identify actual tools that serve as data catalogs and excluded tools whose primary functions do not fall in this category. After filtering, we obtained the 39 tools, shown in Table 4.

Then, we defined the classification framework based on the features we used to filter the tools. Some features were added later or classified differently during an iterative process of filling and validating the framework. The main aspects we considered during the validation and mapping steps are as follows: first, we identified an initial set of high-level capabilities; we next extended this initial set by comparing the

Table 4
List of data catalog tools.

Tool name	Tool ID	License	Type	Deployment
Acryl DataHub	T1	Proprietary	Cloud-native	Cloud, On-Premises
Alation Data Catalog	T2	Proprietary	Platform	Cloud, On-Premises
Alex Augmented Data Catalog	T3	Proprietary	Platform	Cloud
Amazon Web Services Glue Data Catalog	T4	Proprietary	Cloud-native	Cloud
Amundsen	T5	Open Source	Stand-alone	On-Premises
Apache Atlas	T6	Open Source	Stand-alone	On-Premises
Ataccama Data Catalog	T7	Proprietary	Platform	Cloud, On-Premises
Atlan Data Discovery and Catalog	T8	Proprietary	Stand-alone	Cloud, On-Premises
Boomi Data Catalog and Preparation	T9	Proprietary	Platform	Cloud
CKAN	T10	Open Source	Stand-alone	On-Premises
Cloudera Data Catalog	T11	Proprietary	Platform	Cloud, On-Premises
Collibra Data Catalog	T12	Proprietary	Platform	Cloud, On-Premises
Data Galaxy	T13	Proprietary	Platform	Cloud
data.world	T14	Proprietary	Stand-alone	Cloud
Precisely	T15	Proprietary	Platform	Cloud, On-Premises
Dataedo	T16	Proprietary	Stand-alone	On-Premises
DataHub	T17	Open Source	Stand-alone	Cloud, On-Premises
Erwin Data Catalog	T18	Proprietary	Stand-alone	Cloud, On-Premises
Google Cloud Platform Data Catalog	T19	Proprietary	Cloud-native	Cloud
Hygraph	T20	Proprietary	Stand-alone	Cloud
IBM Knowledge Catalog	T21	Proprietary	Stand-alone	Cloud, On-Premises
Informatica Enterprise Data Catalog	T22	Proprietary	Platform	Cloud, On-Premises
K2View	T23	Proprietary	Platform	Cloud, On-Premises
Metacat	T24	Open Source	Stand-alone	On-Premises
Microsoft Purview Data Catalog	T25	Proprietary	Cloud-native	Cloud
Monte Carlo	T26	Proprietary	Stand-alone	Cloud
Octopai	T27	Proprietary	Stand-alone	Cloud
OneTrust	T28	Proprietary	Platform	Cloud, On-Premises
OpenDataDiscovery	T29	Open Source	Stand-alone	On-Premises
OpenMetadata	T30	Open Source	Stand-alone	On-Premises
Oracle Cloud Infrastructure Data Catalog	T31	Proprietary	Cloud-native	Cloud
OvalEdge	T32	Proprietary	Stand-alone	Cloud, On-Premises
Pentaho Data Catalog	T33	Proprietary	Stand-alone	On-Premises
Redgate SQL Data Catalog	T34	Proprietary	Application-specific	On-Premises
Tableau Catalog	T35	Proprietary	Application-specific	Cloud, On-Premises
ThinkData Platform	T36	Proprietary	Stand-alone	Cloud
TIBCO Cloud Metadata (EBX)	T37	Proprietary	Stand-alone	Cloud
Truedat	T38	Open Source	Stand-alone	On-Premises
Zeena	T39	Proprietary	Stand-alone	Cloud

official documentation of tools with the white literature papers, generating a set of fine-grained features; we next mapped this fine-grained features with the high-level ones; we then mapped the features with the metadata life-cycle management steps; we finalized the classification framework with a taxonomy that includes the maturity levels of each tool. The first two authors of the paper collaboratively developed and refined the classification framework and mapping schemes.

3.4. Results

From our MLR, we obtained the following results: (i) a *classification framework with five high-level features and 24 fine-grained features*, (ii) a *list of data catalogs and their comparison based on the classification framework*, (iii) a *schema for mapping features to metadata life-cycle management tasks*, and (iv) a *schema for mapping features to big-data architecture requirements*. Fig. 5 summarizes these artifacts and Sections 4–7–6 present them in detail.

3.5. Replication package

We have made all related data and materials publicly accessible online¹⁴ to facilitate the validation and replication of our research findings. The package includes the complete dataset, detailed steps for qualitative analysis (e.g., categorizations and results), as well as supplementary scripts and tables.

4. RQ1: The classification framework

We identified five high-level and 24 fine-grained features that can be used to classify and compare different data catalog tools. The high-level features reflect the stages that metadata undergo in a data catalog environment, while the fine-grained features represent the atomic capabilities of a data catalog. Fig. 6 shows the features. The five high-level feature categories were derived by clustering fine-grained features based on their functional roles within the metadata lifecycle management, such as data discovery, governance, and collaboration. The process for deriving them is described in Fig. 4. This structure aligns with common metadata management themes in both academic and industry literature we analyzed.

In the rest of this section, we will present each high-level feature and the related fine-grained ones. Additionally, we will present the maturity levels we identified.

4.1. Metadata harvesting

A data catalog represents a collection of metadata related to data assets. Metadata harvesting refers to the process of automatically extracting and indexing these metadata from various data sources (Bianchini et al., 2023; Kobayashi et al., 2018; Hai et al., 2016; Haslhofer and Klas, 2010). The realization of this feature can employ the fine-grained features that support connections to data sources, building a data dictionary, gathering data quality information, and harvesting metadata from a sample of the complete dataset (Yang, 2023; Eichler et al., 2022, 2021; Helal et al., 2021; Cheng-Wei and Bing-Bing, 2018; Beheshti et al., 2018; Bratt et al., 2017; Jett et al., 2016; Cano et al.,

¹⁴ <https://figshare.com/s/1b3a0205006d761ac935>

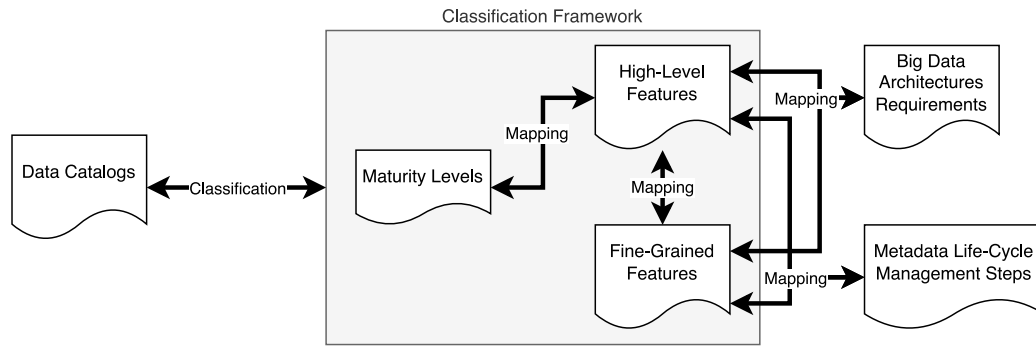


Fig. 5. Summary of classification framework and mappings.

	Metadata Harvesting	Metadata Augmentation	Metadata Usage	Governance	Collaboration
Level 1	Data Source Connectivity Data Dictionary Storage	Documentation Classification	Search Dashboards	Data Lineage Changes Tracking Sensitive Data Detection	
Level 2	Data Sampling Data Quality Information	Business Glossary	Data Profiling	Roles and Responsibilities Task Automation and Standardisation Rules and Policies	Rating Reviewing
Level 3			Data Product	Data Quality Assessment Data Risk	Texting Sharing

Fig. 6. High-level and fine-grained features of data catalog tools, mapped to the data catalog maturity levels.

2023; Grunzke et al., 2019; Amorim et al., 2017; Fox et al., 2006; Sicilia et al., 2019; Milward, 2020; Batwada et al., 2020; Stansberry et al., 2020; Kim et al., 2019).

Data source connectivity (Hai et al., 2021; Bianchini et al., 2023; Shraga and Miller, 2023; Eichler et al., 2022; Helal et al., 2021). An organization generally has different data sources, and a data catalog should be able to connect them to automatically discover, ingest, and index datasets from them (Yang, 2023; Zitoun et al., 2021; Cheng-Wei and Bing-Bing, 2018; Kobayashi et al., 2018; Hai et al., 2016; Iury et al., 2016; Grunzke et al., 2019; Abolhassani et al., 2018). Data sources may store structured, semi-structured, and unstructured data and use various data storage types, including SQL, NoSQL (not only SQL), and file storage (Bianchini et al., 2023). Data catalogs typically provide connectors that can automatically or semi-automatically collect metadata from various data sources (Bianchini et al., 2023; Hai et al., 2021; Pingos and Andreou, 2023; Schrott et al., 2023; Shraga and Miller, 2023; Holl and Gossling, 2019; Kobayashi et al., 2018; Hai et al., 2016; Iury et al., 2016; Haslhofer and Klas, 2010). Connectors may be augmented with AI to extract metadata from unstructured data sources, infer data asset schema, capture metadata such as entities, data types, and relationships, and automate data crawling processes.

Data dictionary (Eichler et al., 2022; Holl and Gossling, 2019; Li and Sugimoto, 2017; Iury et al., 2016; Shastri et al., 2020). The IBM Dictionary of Computing (McDaniel, 1994) defines *data dictionary* as:

“A centralized repository of information about data such as meaning, relationships to other data, origin, usage, and format”.

A data dictionary serves as a reference that defines and explains the structure and content of the data, making it easier for users to understand and utilize the data correctly (Eichler et al., 2021). For example, Schrott et al. (2023) used metadata derived from data dictionaries to build an ontology layer for data catalogs, helping to describe and relate data objects semantically.

Storage (Travkin, 2024; Ehrlinger et al., 2021). Data catalogs store metadata in a backend database, which can be relational, NoSQL, or graph-based. Usually, the backend is configurable, allowing organizations to choose databases that align with their infrastructure. They manage metadata schemas, relationships, and lineage, often stored as SQL tables, JSON, or RDF. In some cases, it is possible to find hybrid architectures, combining databases for scalability and flexibility.

Data sampling (Yang, 2023; Eichler et al., 2022; Holl and Gossling, 2019; Iury et al., 2016). This feature enables users to preview a data asset – usually a small and representative subset of it – from which metadata have been collected by the catalog. It allows users to quickly understand the structure, content, and characteristics of large datasets before fully integrating them into workflows.

Data quality information (Yang, 2023; Sousa et al., 2023; Sawadogo and Darmont, 2021b; Bratt et al., 2017; Hai et al., 2016). This feature

indicates the tool's capability of displaying data quality metrics of an asset that generated from the source and are collected at ingestion time together with other metadata (Kumar et al., 2024; Eichler et al., 2022, 2021; Aggour et al., 2017; Sinthong et al., 2021; Cherradi et al., 2023). In this case, the catalog is not capable of evaluating these metrics. Catalogs incorporate quality information in several ways. For instance, data quality scores are assigned at the table level for each data asset and evaluated over pre-defined data quality rules on the ingested metrics. Some examples are rules for checking if the percentage of missing values is above a given threshold (Kumar et al., 2024; Wang et al., 2016).

4.2. Metadata augmentation

The metadata collected by catalogs might be incomplete or require to be augmented with additional information. A data source might not provide quality metrics, such as integrity, accuracy, authenticity, and uniqueness (Yang, 2023). A data asset might lack the documentation of its data models, data semantics, and business context (Zitoun et al., 2021; Sawadogo and Darmont, 2021b; Abolhassani et al., 2018; Batwada et al., 2020). Metadata augmentation allows the enrichment of metadata with additional information, such as semantics, labels, annotations, and quality information (Hai et al., 2021; Bianchini et al., 2023; Kumar et al., 2024; Papenbrock et al., 2015; Bratt et al., 2017; Maccioni and Torlone, 2017; Hai et al., 2016; de Oliveira et al., 2022; Eichler et al., 2022; Sawadogo and Darmont, 2021b; Grunzke et al., 2019; Abolhassani et al., 2018; Boukhers et al., 2023). Catalog features, such as business glossaries and classification, enable enriching data assets by incorporating tags and labels based on criteria such as asset type, sensitivity level, business-criticality, quality score, and business terms adopted by an organization (e.g., business domain, data domain, and department) (Eichler et al., 2022).

Data documentation (Shraga and Miller, 2023; Eichler et al., 2022; Zitoun et al., 2021; Eichler et al., 2021). The documentation of a data asset typically includes brief descriptions of data, links to other resources, such as glossary terms, dashboards and related datasets, and references to documentation outside the catalog, such as APIs documentation, ETL pipelines documentation and reports. Data catalogs may employ AI in managing documentation, for example, generating documentation by automatically summarizing and interpreting metadata attributes (Holl and Gossling, 2019; Kobayashi et al., 2018; Bratt et al., 2017; Hai et al., 2016; Milić et al., 2021). Many catalogs use built-in Markdown or Rich Text editors to create and store documentation or employ Wiki-based and graph-based technologies, e.g., knowledge graphs and graph databases (such as Neo4j¹⁵).

Classification (Eichler et al., 2022; Beheshti et al., 2018; Vries et al., 2022; Grunzke et al., 2019; Cherradi et al., 2023; Li and Sugimoto, 2017; Aggour et al., 2017; Bratt et al., 2017; Hai et al., 2016; Cano et al., 2023; Kobayashi et al., 2018; Milić et al., 2021; Shanmugam and Seshadri, 2016). This feature refers to the process of organizing and classifying data assets based on some criteria that span from data type and sensitivity to usage and quality. This feature helps users identify data assets based on their characteristics (e.g., specific data class or business term). Catalogs may offer ML-based or rule-based classification capabilities that scan and categorize assets based on their content (Sousa et al., 2023; Boukhers et al., 2023; Stansberry et al., 2020). The classification process typically adds tags and labels to data sets, enabling advanced search capabilities. Catalog users can also personalize their experience by assigning personal labels to data assets.

Business glossary (Bianchini et al., 2023; Shanmugam and Seshadri, 2016; Eichler et al., 2021, 2022). The business glossary is a centralized repository of agreed-upon definitions and descriptions of business terms. It represents a bridge between business and technical terms and can ensure that everyone in an organization understands and uses the same vocabulary when interpreting and analyzing data. To formally define the terms without ambiguity and capture complex relationships among them, data catalogs may use knowledge graphs/ontologies (Schrott et al., 2023; Seabolt et al., 2018; Cherradi et al., 2023).

4.3. Metadata usage

Once metadata are collected, stored, and augmented, they can be used. Metadata users can be either a software application (e.g., the data discovery component of a data catalog) or a human (Pingos and Andreou, 2023; Vries et al., 2022). A common use case for metadata is developing dashboards for data quality visualization, data asset usage monitoring, or data lineage visualization (Cherradi et al., 2023; Sousa et al., 2023; Shraga and Miller, 2023; Huang, 2020; Li and Sugimoto, 2017; Cronholm et al., 2017; Seabolt et al., 2018; Armbrust et al., 2020; Shastri et al., 2020; Abolhassani et al., 2018; Shanmugam and Seshadri, 2016). Catalogs might offer the feature to define data products directly in the catalog by collecting and organizing metadata accordingly (Eichler et al., 2021, 2022).

Search (Eichler et al., 2022, 2021; Helal et al., 2021; Holl and Gossling, 2019). The search feature enables the discovery of data assets. Data catalogs implement this feature with different technologies: full-text search engines, like Elasticsearch¹⁶; Natural Language Processing (NLP); search ranking algorithms; knowledge graphs, which are at the basis of the data discovery platform proposed by Helal et al. (2021), enable to find relationships between datasets and efficiently integrate them. The type of technology that forms the basis for the search engine used by a data catalog has a huge impact on data discovery and its impact once adopted, especially if enhanced by AI, capable of providing insights and personalized recommendations suggest related datasets based on user activity and metadata context (Beheshti et al., 2018; Cherradi et al., 2023).

Dashboards (Sousa et al., 2023). Dashboards provide visual insights into data usage, governance status, lineage, and quality metrics. Metadata also power data lineage and impact analysis by tracking dependencies across datasets, and supports data quality monitoring through automated checks and completeness or accuracy scoring. Visualizing metrics, insights, and any information related to data assets is a feature that can be found in catalogs.

Data profiling (Liu and Zsu, 2009; Zitoun et al., 2021; Helal et al., 2021; Maccioni and Torlone, 2017; Papenbrock et al., 2015; Grunzke et al., 2019). Data profiling generates statistics or representative summaries of a data asset to understand its structure and quality. Data catalogs generally can perform this activity when extracting metadata from a dataset.

Data product (Dehghani and Fowler, 2022; Goedegebuure et al., 2024). Dehghani (Dehghani and Fowler, 2022) introduces data as a product as the second principle of the data mesh architecture.

[It] applies product thinking to domain-oriented data to remove usability frictions and truly delight the experience of data users.

The desired attributes of data products are discoverability, addressability, understandability, trustworthiness, native accessibility, interoperability, independent value, and security. Some catalogs provide built-in

¹⁵ <https://neo4j.com/>

¹⁶ <https://www.elastic.co/elasticsearch>

features for storing the metadata of data products and monitoring and regulating access to data products, enabling achieving properties such as discoverability, understandability, and security (Eichler et al., 2022; Dolhopolov et al., 2023; Goedegebuure et al., 2024).

4.4. Governance

Abraham et al. (2019) analyzed the existing literature on data governance, including various definitions for data governance, and created a common definition.

“Data governance specifies a cross-functional framework for managing data as a strategic enterprise asset. In doing so, data governance specifies decision rights and accountabilities for an organization’s decision-making about its data. Furthermore, data governance formalizes data policies, standards, and procedures and monitors compliance”.

Metadata are vital inputs for governance operations, as they provide observability and monitoring of the operations on data happening across the organization (Yang, 2023; Pingos and Andreou, 2023; Zitoun et al., 2021; Sawadogo and Darmont, 2021b; Cronholm et al., 2017; Armbrust et al., 2020; Amorim et al., 2017). However, they can also be the outputs of governance operations, such as data policies and the roles and responsibilities of stakeholders (Sousa et al., 2023; de Oliveira et al., 2022; Eichler et al., 2022; Seabolt et al., 2018; Shastri et al., 2020; Shanmugam and Seshadri, 2016; Dong et al., 2016; Casanovas et al., 2016). Monitoring and recording how data assets are accessed, used, and modified is a feature that some catalogs offer. This can be implemented at different levels, such as log monitoring, audit trails, and access patterns. Information collected from the consumers who use data assets can be exploited to understand their value and impact on an organization. For instance, if a data catalog can collect data quality metrics, it might also automatically monitor if metric values are inconsistent and thus trigger alerts or initiate workflows for handling or reporting quality issues. Another automated element is to assign tasks to individuals with specific roles, such as data stewards, to investigate and resolve issues. For instance, a catalog might track how a data asset is used in multiple downstream dashboards and machine learning models, allowing users to assess the impact of changes to the dataset on business processes (Hai et al., 2021; Sousa et al., 2023; Shastri et al., 2020; Tang et al., 2019; Abolhassani et al., 2018). We identified several fine-grained features that describe the various aspects of governance: data lineage, change tracking, sensitive data detection, roles and responsibilities, task automation and standardization, rules and policies, data quality assessment, and data risk (Dolhopolov et al., 2023; Cherradi et al., 2023; Muñoz et al., 2021; Batwada et al., 2020; Stansberry et al., 2020).

Data lineage (Hai et al., 2021). Data lineage is the process of collecting, recording, and visualizing the flow of data from its sources to destinations. Data catalogs offer this feature, sometimes enhanced by AI, for example, augmenting lineage by completing missing information or adding tags and semantic descriptions to lineage steps (Schrott et al., 2023; de Oliveira et al., 2022; Eichler et al., 2022; Huang, 2020; Holl and Gossling, 2019; Maccioni and Torlone, 2017; Li and Sugimoto, 2017; Aggour et al., 2017; Hai et al., 2016; Shastri et al., 2020; Tang et al., 2019; Abolhassani et al., 2018; Shanmugam and Seshadri, 2016; Guedes et al., 2020; Interlandi et al., 2018; Cherradi et al., 2023; Dang and Anh, 2021).

Changes tracking (Shraga and Miller, 2023; de Oliveira et al., 2022; Eichler et al., 2022, 2021; Huang, 2020; Armbrust et al., 2020). This feature refers to the capability to monitor and record modifications made to data assets and their metadata over time. It helps an organization maintain transparency and control over its data assets. Capabilities such as version control, logging, and visualizations support realizing this feature.

Sensitive data detection. The ease of identifying sensitive data in data sets can help organizations to accurately implement the procedures for protecting personal data. Cataloging tools provide built-in capabilities to find and label sensitive data explicitly, for example, by assigning a specific tag or flag to data fields containing sensitive information, such as social security numbers or health information (Aggour et al., 2017; Shastri et al., 2020; Cherradi et al., 2023).

Roles and responsibilities (Pingos and Andreou, 2023; Sawadogo and Darmont, 2021b; Cronholm et al., 2017; Seabolt et al., 2018; Shastri et al., 2020; Shanmugam and Seshadri, 2016; Dong et al., 2016; Dolhopolov et al., 2023; Stansberry et al., 2020). A data governance structure typically consisted of a set of roles such as data stewards, data owners, domain owners, data scientists, and data platform administrators (Eichler et al., 2022). Each role may be given specific permissions for viewing, editing, and managing assets. Data catalog tools should support defining these roles and their responsibilities. This feature ensures that data assets and metadata are properly governed and maintained by the right individuals. In data governance workflows, such as reviewing and approving data, appropriate responsibilities can be assigned to specific roles to ensure the right stakeholders are notified. Also, governance policies (e.g., data protection and retention) can be defined and enforced so that only specific roles can perform particular tasks (e.g., accessing sensitive data).

Task automation and standardization (Yang, 2023; Shraga and Miller, 2023; Eichler et al., 2022; Cronholm et al., 2017; Seabolt et al., 2018; Shastri et al., 2020; Vries et al., 2022; Dong et al., 2016; Dolhopolov et al., 2023; Stansberry et al., 2020). Data catalogs may support automating tasks such as creating and reviewing data governance policies, granting data access, and creating triggers and templates in a standardized way. Enforcing consistent and standardized processes for managing data assets can reduce manual errors, improve compliance, and enhance efficiency.

Rules and policies (Eichler et al., 2022; Cronholm et al., 2017; Sawadogo and Darmont, 2021b; Hai et al., 2016; Seabolt et al., 2018; Milić et al., 2021). Some tools provide features to define rules and policies to enforce data governance and compliance. For instance, data quality rules, data integrity, retention (e.g., triggering deletion workflows once data reaches the end of its life-cycle), usage of data (Shanmugam and Seshadri, 2016; Dolhopolov et al., 2023; Cherradi et al., 2023; Stansberry et al., 2020) and compliance rules (e.g., GDPR compliance) (Shastri et al., 2020; Dong et al., 2016) can be enforced and automated within the data catalog. Data assets that violate these rules are flagged, facilitating the identification and review of the problematic data.

Data quality assessment (Yang, 2023; Sousa et al., 2023; Hai et al., 2016; Eichler et al., 2021, 2022; Cherradi et al., 2023). Assessing the quality of a dataset involves measuring a set of quality dimensions such as accuracy (a measure of the degree to which data correctly reflects real-world events or objects), uniqueness (a measure of a fraction of duplicate data), and completeness (a measure of a fraction of missing data) (Ehrlinger and Wöß, 2022; Ustunboyacioglu et al., 2024). Data catalog tools should be able to compute data quality metrics on the data they manage and store the calculated metrics as additional metadata (Hai et al., 2021; Kumar et al., 2024; Sawadogo and Darmont, 2021b; Sinthong et al., 2021; Abolhassani et al., 2018; Boukhers et al., 2023; Wang et al., 2016). This evaluation is performed by applying quality checks and rules, and automatically flagging quality issues if found. For instance, completeness checks can be performed when crawling data sources to ensure no duplicate rows exist.

Data risk (Hai et al., 2021; Yang, 2023; Aggour et al., 2017). This feature refers to analyzing and assessing the risks associated with data assets, including compliance risks, regulation risks, security risks, and quality issues (Hai et al., 2021; Yang, 2023; Aggour et al., 2017;

Shastri et al., 2020; Abolhassani et al., 2018; Wang et al., 2016). Data catalogs may offer components that help identify such risks, generate reports, and remediate risks, usually by integrating them with data risk management tools. For instance, a catalog could provide insights on who accessed a database containing sensitive data by monitoring access patterns, or a lineage graph could show that sensitive data are shared with a third-party organization, highlighting potential risks with third-party data sharing. Also, data catalogs can help ensure data are compliant with regulations such as GDPR by enforcing governance rules (Cronholm et al., 2017; Eichler et al., 2022).

4.5. Collaboration

Many data catalogs allow the various stakeholders of metadata to interact with each other and provide feedback on the data assets available on the catalog (Eichler et al., 2022; Shanmugam and Seshadri, 2016; Stansberry et al., 2020; Eichler et al., 2021). For instance, a data consumer can comment on a data asset's quality, and the data provider can check the comment and act accordingly. Stakeholders, such as data users and business analysts, can make the data catalog and stored assets more valuable and reliable by rating and commenting on data assets, actively engaging with data assets owners, data stewards, and other actors via integrated chat services, and sharing documentation about data assets (Bratt et al., 2017; Anon, 2024d). This process allows all the users to contribute to managing the data catalog and providing proof of the benefits of its adoption (de Oliveira et al., 2022; Iury et al., 2016; Parra-Moyano et al., 2024).

Texting (Eichler et al., 2022; Shanmugam and Seshadri, 2016). The texting feature of a data catalog allows users to send text-based messages within a data catalog environment. It can enhance real-time interaction between catalog users.

Reviewing (Eichler et al., 2022; Iury et al., 2016; Shanmugam and Seshadri, 2016). This feature allows users to write detailed evaluations, comments, or critiques about data assets. Compared with rating, it enables users to provide qualitative feedback.

Rating (Eichler et al., 2022; Shanmugam and Seshadri, 2016). The rating feature allows users to rank data assets within a catalog quantitatively, typically using a 5-point scale. It is a community-driven metric for assessing a data asset's perceived value and quality.

Sharing (Eichler et al., 2022, 2021; Stansberry et al., 2020). Some data catalogs allow users to share datasets via directly accessible links, making it easy to provide access to data quickly. These links can often be set with different levels of access permissions. Users can collaborate more effectively with colleagues on data projects by enabling direct data sharing.

4.6. Maturity levels

To better understand the evolution of metadata management capabilities in contemporary data ecosystems, we propose a model with three maturity levels for data catalogs. These levels are designed to reflect the progressive enrichment of features, aligned with emerging use cases in data mesh, data lakehouse, and data lake architectures (Travkin, 2024; Corcho et al., 2024). Importantly, these levels are not intended to measure tool reliability or adoption, but rather to categorize tools based on functionality and alignment with modern metadata needs.

As illustrated in Fig. 6, each level corresponds to a set of capabilities that build upon one another. Level 1 represents foundational metadata harvesting and visibility. At this stage, data catalogs focus on indexing and exposing datasets, typically through basic features like connectivity to data sources, a standard data dictionary, and simple search functionality. These capabilities enable organizations to take initial steps

towards centralizing metadata and making it discoverable, especially for technical users.

Level 2 introduces more advanced metadata management capabilities focused on data governance and collaborative use cases. Catalogs at this level provide richer context around data assets by incorporating business glossaries, profiling capabilities, and data quality indicators. They also address organizational governance requirements such as role-based permissions, lineage tracking, and standardized definitions. This maturity level also facilitates improved communication and trust between business and technical stakeholders.

Level 3 marks the transition to strategic business value creation from enterprise data, where metadata becomes the foundation for data productization and advanced data governance. Catalogs at this level allow organizations to treat data as a product, offering features such as curated and versioned data packages, collaborative tools for users to annotate or comment, and mechanisms to assess data quality or privacy risks. Integration with internal or external data marketplaces becomes possible, enabling governed data exchange and monetization. This maturity level supports scalable, decentralized architectures and reflects a high level of automation and user engagement: it is well-aligned with the principles of data mesh, as it supports federated governance and promotes shared understanding of data across domains (Dehghani and Fowler, 2022).

The proposed maturity levels resonate with broader maturity frameworks found in the literature. For instance, Immuta's Data Marketplace Maturity Model (Immuta, 2025) identifies a trajectory from basic access control towards optimized, monetizable data services. Similarly, Corcho et al. (2024) propose a maturity model for semantic artifact catalogs and provide a detailed breakdown of 12 dimensions, similar to our classification framework. Their model aims to help evaluate catalog readiness for supporting semantic interoperability in complex research infrastructures. Their structured, multi-dimensional approach aligns conceptually with our level-based perspective.

5. RQ2: Tools for data cataloging

In this section, we classify 39 data catalog tools we identified from our MLR (see Table 4), using the framework we presented in the previous section.

Fig. 7 provides general information about the tools. We identified four categories of tools: cloud-native, platform-based, standalone, and application-specific. Cloud-native tools such as AWS Glue Catalog and Microsoft Purview Data Catalog are designed for seamless operation in cloud ecosystems, leveraging scalability and integration. Platform-based tools like Informatica Enterprise Data Catalog offer comprehensive solutions, integrating cataloging, governance, and analytics within enterprise architectures. They are typically provided as services/components within an existing data platform solution. Standalone tools such as DataHub, Apache Atlas, and OpenMetadata focus solely on metadata management or cataloging, providing flexibility and interoperability. Application-specific solutions (e.g., Tableau Catalog) focus on specific applications or frameworks, catering to their unique metadata and governance needs. Almost 80% of the tools are proprietary. However, we noticed that 26% of the proprietary solutions are built on open-source tools. For example, Acryl DataHub (proprietary) uses DataHub (open source). Finally, we observed that application-specific ones become generic with the passing years as they are typically extended to support emerging use cases and technologies.

Figs. 8 and 9 illustrate the extent to which various features are supported by data catalog tools. Features related to metadata harvesting (e.g., data source connectivity, storage, and data quality information) and governance (e.g., data lineage, sensitive data detection, roles and responsibilities, and data risk) exhibit widespread support, on average more than 75% adoption. This strong adoption likely reflects their critical role in metadata collection, data quality assurance, security, and compliance. For instance, Collibra (Collibra Documentation, 2024)

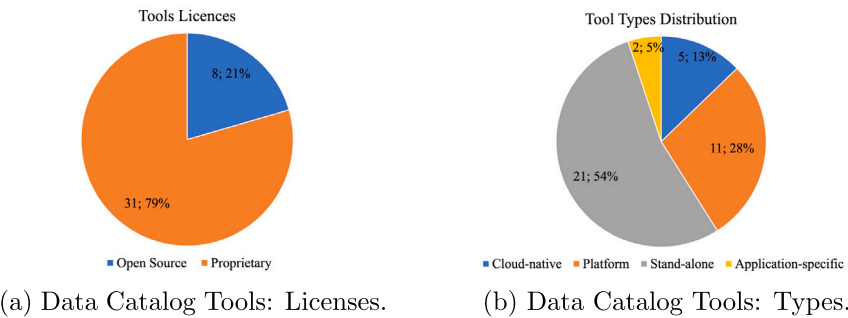


Fig. 7. Data catalog tools general information.

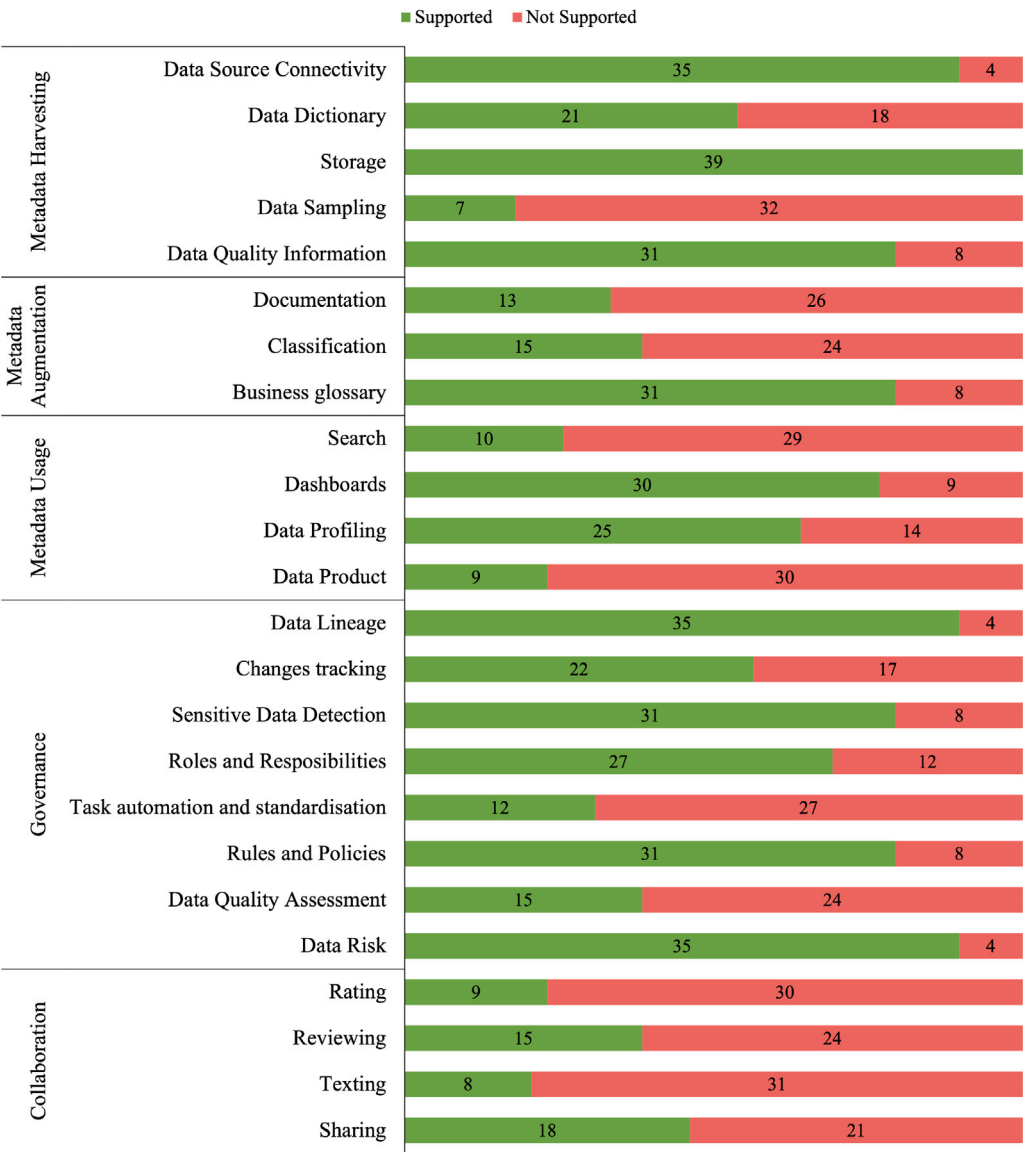


Fig. 8. Features supported by data catalogs. It compares the support for various features across 39 data catalog tools, indicating whether each feature is supported. The bars are split into two colors: green for *Supported* features and red for *Not Supported* features. Each feature is listed on the y-axis, with the number of tools supporting or not supporting it on the x-axis. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

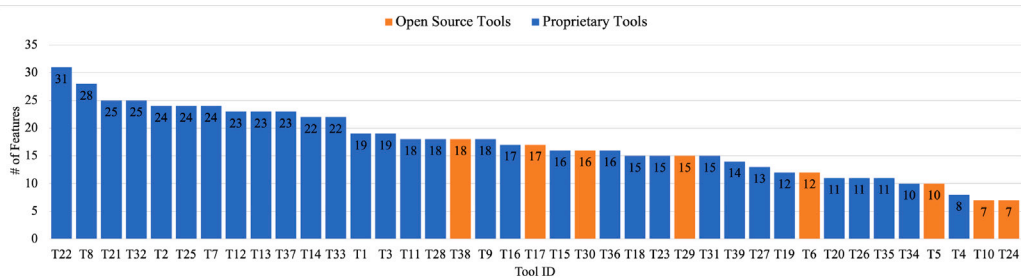


Fig. 9. Ranking of tools' features support.

integrates data quality rules, assigning quality flags to assets with issues. Meanwhile, features such as data dictionary, data usage tracking, and data profiling receive moderate support. This suggests that while governance and monitoring remain key priorities, standardization and automation are still areas for improvement.

In contrast, collaboration-focused features, such as texting, sharing, reviewing, and rating, less than 25% adoption. This indicates that many data catalogs may not prioritize user interaction, feedback, or collaborative metadata management workflows. For instance, *Atlan* (2024) integrates Slack¹⁷-like messaging within its data catalog, enabling users to tag teammates and discuss metadata or quality issues in real time. Additionally, task automation and standardization, as well as data product support, remain limited. This may be attributed to the relative immaturity of the data product concept and the inherent complexity of automating metadata-related workflows.

Despite being a core capability, search functionality is supported by only 10 out of 39 tools (26%), as shown in Fig. 8. This limited adoption is surprising, given that search is essential for data discovery and usability in data catalog environments. However, even if not fully implemented as a dedicated or advanced feature, many tools still provide basic mechanisms for searching data assets, often through keyword filtering or structured queries. Among those with more advanced capabilities, Zeenea offers natural language search (NLP), allowing users to enter prompts that are translated into filters and queries (Zeenea, 2023). Alation enhances user experience through search ranking, surfacing the most relevant or frequently accessed assets using metadata-driven relevance scores (Rahmani, 2021). CoreKG, on the other hand, integrates full-text and semantic search, combining Elasticsearch with SPARQL and SQL querying, and enriches data using external knowledge bases (Beheshti et al., 2018).

As we can observe in Fig. 9, open-source tools support fewer features, highlighting a potential gap in comprehensiveness compared to proprietary solutions. 5 tools support 10 or fewer features, and 3 of them are open-source. The top 10 tools are proprietary, and the largest number of supported features by a single tool is 31. On average, each tool supports 16 features. There is a noticeable gap between tool comprehensiveness: a difference of 24 features between the most and least capable tools (31 vs. 7).

Table 5 provides insights on how each feature is implemented in 5 open source tools, while Table 6 provides such information for 5 proprietary tools. These tables aim to help the reader contextualize each feature with real-world examples.

Overall, data catalogs emphasize connectivity, documentation, and quality management, yet they lag in collaboration, automation, and user engagement. As data governance continues to evolve, future catalog solutions may place greater emphasis on these currently under-supported features.

Table 7 presents an overview of 39 data catalogs, each classified according to its maturity level. The most common category is *Level*

1, which suggests that many tools are designed to help organizations maintain a structured and well-managed set of critical data assets. The tools in *Level 2* are tailored for organizations that need to enforce data governance policies and ensure regulatory compliance. The tools in *Level 3* focus on facilitating collaboration among data catalog stakeholders, ensuring data quality, and supporting the discovery, sharing, and trading of data assets as data products within an organization.

6. RQ3: Mapping the data catalog features to metadata life-cycle management steps

Metadata life-cycle management is the process of creating, storing, updating, and retiring metadata. Academic and grey literature provides several models for data and metadata life-cycle management (Sinaeepourfard et al., 2016a; Holom et al., 2020; Potvin and Thompson, 2016; Government of Canada, 2024; Anon, 2025; Barton and Robertson, 2005; Sinaeepourfard et al., 2016b; Eichler et al., 2022, 2021). In this section, we consider the model proposed in Anon (2025), Government of Canada (2024), which reflects the findings from academic literature (Potvin and Thompson, 2016; Eichler et al., 2021; Holom et al., 2020; Sinaeepourfard et al., 2016a). We use it to assess the extent to which data catalogs support metadata life-cycle activities.

Fig. 10 summarizes the steps in the adopted metadata lifecycle management workflow (Anon, 2025; Government of Canada, 2024; Potvin and Thompson, 2016; Sinaeepourfard et al., 2016a). This process aims to optimize data/information understanding, maximize its usage, and reduce time to manage it. There are eight phases in the process.

Plan. The planning phase is crucial for understanding metadata requirements and identifying key stakeholders. This involves analyzing and documenting metadata-related challenges within the organization, reviewing existing data governance roles and processes, and formulating a strategy for implementing a metadata management solution. Additionally, this stage involves reviewing established metadata schemas to ensure their compatibility with business needs.

Construct, Create, Acquire. This phase involves selecting metadata schemas and defining technical requirements. It is important to acquire a data dictionary and business glossary to standardize metadata terminology. A key objective is to establish agreements on how metadata will be created, maintained, integrated, and accessed. Identifying potential risks and issues is also part of this phase.

Commission, Organize, Store. Metadata architecture and governance frameworks are established in this stage, defining organizational roles, responsibilities, and quality assurance processes. Metadata are collected from various sources, classified based on sensitivity and criticality, and stored in a structured manner. This stage also involves documenting policies and procedures related to metadata governance. Collaboration with relevant business units ensures alignment with organizational objectives.

Access. This stage ensures that authorized users and applications can access metadata efficiently. It includes providing tools for harvesting, publishing, interpreting, and sharing metadata. The primary outcome is the establishment of standardized methods for accessing and

¹⁷ <https://slack.com/>

Table 5

Open source tools features implementations. Atlan, Purview, and Cloudera are built on Apache Atlas, while Acryl DataHub is built on Datahub. RBAC (Role-based Access Control), SSO (Single Sign-On), LDAP (Lightweight Directory Access Protocol).

		Amundsen	Apache Atlas	DataHub	Open data discovery	Open metadata
Metadata Harvesting	Data Source Connectivity	Push/pull from structured, semi-structured sources	Push from structured, semi-structured sources	Push/pull from structured, semi-structured, unstructured sources	Push/pull from structured, semi-structured, unstructured sources	Push/pull from structured, semi-structured sources
	Data Dictionary	Supported	Not Supported	Supported	Supported	Supported
	Storage	Neo4j, RDS	JanusGraph ^a , Hbase ^b , MySQL ^c , PostgreSQL	MySQL, PostgreSQL	PostgreSQL	MySQL
	Data Sampling	Ingestion Supported	Not Supported	Not Supported	Not supported	Ingestion Sampling
	Data Quality Information	Integrates with external tools	Third-party integrations	Data Quality Assertions	Data Quality Tests	Data Quality Tests
Metadata Augmentation	Documentation	Markdown-based	Descriptions, Data types	Descriptions, Data types, Connectors for documentation	Descriptions, data types	Connectors, entities for documentation
	Classification	Tags	Apache Ranger ^d , Tags	Tags, Data Types	Tags and Labels	Tags, Tiers
	Business Glossary	Glossary Terms	Business Taxonomy	Glossary Terms	Dictionary Terms	Glossary
Metadata Usage	Search	Elasticsearch, Neo4j (Index)	Apache Solr ^e	Elasticsearch, Neo4j (Index)	Elasticsearch	Elasticsearch
	Dashboards	Third-party integrations.	Supported	Supported	Data Quality	KPIs, Analytics, Cost Analysis
	Data Profiling	Not Supported	Not Supported	SQL Profiling	Great Expectations ^f , Pandas ^g	Data Profiler UI, Workflow
	Data Product	Not Supported	Not Supported	Supported as UI and Attributes	Not supported	Metadata Schema
Governance	Data Lineage	Supported	Supported	Column-level Lineage	End-to-end lineage	Column-level Lineage
	Changes Tracking	Metadata Versioning	Metadata Versioning	Metadata Versioning	Schema Diff	Data Asset Versioning
	Sensitive Data Detection	Flags for GDPR, CCPA, HIPAA, etc	Apache Ranger	Tags	Not supported	Tags
	Roles and Responsibilities	Role-based management, LDAP and IAM integration	Authentication and Authorization	RBAC, Roles, Permissions, Privileges	User-owner association, Roles, Permissions	RBAC, SSO, Authentication
	Task Automation and Standardization	Automated Classification, Airflow	Hooks, Classification Propagation, Airflow	Auto-classification, Discovery, Transformations	Workflows, Airflow	Auto-classification, Workflows, Alerting, Airflow
	Rules and Policies	RBAC, LDAP.	Apache Ranger	Platform and Metadata Policies	RBAC	Rules, Conditions, Policies
	Data Quality Assessment	Third-party Integrations	Third-party Integrations	Third-party Integrations	Third-party Integrations	Third-party Integrations
	Data Risk	PII, Financial	PII, Sensitive	PII	PII	PII, Verified
Collaboration	Rating	Not Supported	Not Supported	Not Supported	Not Supported	Not Supported
	Reviewing	Not Supported	Not Supported	Not Supported	Not Supported	Requests, Announcements, alerts
	Texting	Slack and Teams integration	Not Supported	Not Supported	Slack Events API	Conversation threads
	Sharing	REST APIs	REST APIs	Not Supported	Not Supported	Not Supported

^a <https://janusgraph.org/>^b <https://hbase.apache.org/>^c <https://www.mysql.com/>^d <https://ranger.apache.org/>^e <https://solr.apache.org/>^f <https://greatexpectations.io/>^g <https://pandas.pydata.org/>

Table 6
Proprietary tools: Features implementation.

Category	Feature	AWS glue catalog	Tableau catalog	IBM knowledge catalog	Collibra data catalog	Alation
Metadata Harvesting	Data Source Connectivity	AWS Glue	Tablau only	Push/Pull from structured, semi-structured, unstructured sources	Push/pull from structured, semi-structured, unstructured sources	Push/pull from structured, semi-structured, unstructured sources
	Data Dictionary	Not supported	Supported as UI	Supported as UI	Supported as UI and Attributes	Supported
	Storage	AWS S3, AWS Redshift ^a , AWS RDS, etc.	PostgreSQL	PostgreSQL, metadata graph	PostgreSQL	PostgreSQL, KVStore ^b , RocksDB ^c
	Data Sampling	Not supported	Not supported	Not supported	Sampling at metadata ingestion with Jobserver	Supported at ingestion
	Data Quality Information	Not supported	External tools	Data Quality rules	Integration with 3rd party tools	Data Quality Processor
Metadata Augmentation	Documentation	Descriptions, Data types	Descriptions	Descriptions, Data types	Descriptions, Data types	Document Hub
	Classification	AWS Lake Formation ^d Tags	Tags	Auto-tagging via Watson AI ^e	Tags	Tags
	Business Glossary	AWS Lake Formation Tags	Not Supported	Business Terms	Glossary Terms	Glossary Terms
Metadata Usage	Search	AWS Athena, AWS Redshift	Proprietary search index	AI-powered search (Watson NLP)	Elasticsearch	Elasticsearch, AI support
	Dashboards	Not supported	Supported	Supported	Supported	Supported
	Data Profiling	Not supported	Not Supported	Supported	Supported via REST API	Supported
	Data Product	Not supported	Not Supported	Not supported	Supported	Not Supported
Governance	Data Lineage	Partial (with AWS Lake Formation)	Tableau only lineage	Graph-based lineage	End-to-end lineage	Column-level lineage
	Changes Tracking	Assets Versioning	Assets Versioning	Assets Versioning	Assets Versioning	Asset versioning
	Sensitive Data Detection	Sensitive Data Detection	Tags	Tags	Tags	Tags, Sensitive data
	Roles and Responsibilities	AWS IAM/Lake Formation	RBAC	RBAC and attribute-based	RBAC	RBAC, ACLs
	Task Automation and Standardization	Triggers, AWS Glue Workflows	Workflows	Powered by Watson AI: metadata enrichment, smart suggestions, NLP, automatic classification	No built-in AI, workflows	Celery, workflows
	Rules and Policies	AWS Lake Formation	Access Rules	Policy engine (access, quality, privacy), policy lifecycle management	BPMN-based workflow engine (Activiti), custom policies	Access Rules, Quality Rules, Policies, Permissions
	Data Quality Assessment	Not Supported	Not Supported	Scorecards, anomaly detection	Data Quality processor	
	Data Risk	AWS Macie ^f	Tags	PII detection and classification	Manual Tags	PII Tags
Collaboration	Rating	Not Supported	Not supported	Supported	Supported	Not Supported
	Reviewing	Not Supported	Supported	Supported	Supported	Not Supported
	Texting	Not Supported	Not supported	Not supported	Not supported	Not Supported
	Sharing	REST APIs	Not supported	Not supported	Not supported	Not Supported

^a <https://aws.amazon.com/redshift/>

^b <https://www.kvstore.io/>

^c <https://rocksdb.org/>

^d <https://aws.amazon.com/lake-formation/>

^e <https://www.ibm.com/watson>

^f <https://aws.amazon.com/maciek/>

integrating metadata into business processes. Additionally, this phase promotes metadata reuse and indexing, improving discoverability.

Use. The primary focus of this phase is operationalizing metadata. Key activities include training staff on metadata management, reviewing metadata policies, monitoring metadata usage, and generating reports. Effective governance is reinforced by tracking how metadata are utilized across the organization.

Assess. Assessment of metadata quality and governance is performed from both business and technical perspectives. This includes adhering to metadata quality standards, maintaining metadata consistency, and safeguarding sensitive information. Additionally, the harmonization and integration of metadata across different systems are evaluated. Governance processes are in place to ensure compliance with organizational policies.

Table 7
Maturity levels of data catalog tools.

Maturity level	Number of tools	Proprietary tool IDs	Open source tool IDs
Level 1	22	T1, T2, T4, T11, T13, T14, T15, T16, T18, T20, T26, T27, T31, T33, T34, T35, T37	T5, T6, T17, T24, T29
Level 2	11	T3, T7, T9, T19, T21, T23, T25, T28, T32, T36	T30
Level 3	6	T8, T12, T22, T39	T10, T38

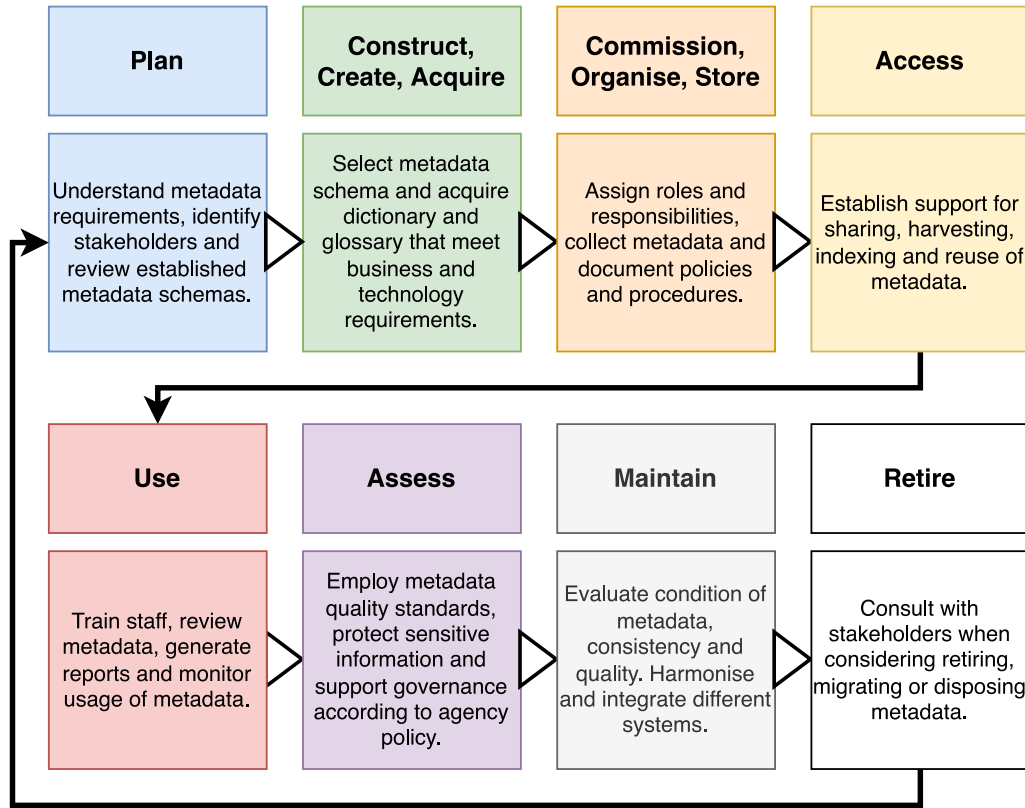


Fig. 10. Metadata life-cycle management steps.

Maintain. Metadata maintenance is enhanced through automated processes, including tagging, harvesting, and quality checks. Organizations can leverage glossaries, data dictionaries, and lineage tracking to improve metadata consistency and usability. Integration with business processes further enhances the value of metadata by ensuring continuous updates and quality improvements.

Retire. The retirement phase focuses on managing obsolete or redundant metadata. Metadata may become obsolete due to changing business requirements or the deprecation of associated data. This phase involves consulting stakeholders to determine whether metadata should be retired, migrated, or archived. Additionally, metadata migration considerations, such as preserving metadata lineage and ensuring proper archiving, should be addressed to maintain historical records.

Table 8 maps high-level and fine-grained features to metadata life-cycle management steps, highlighting their role in effectively managing metadata. A key observation is the consistent emphasis on governance and metadata augmentation across most lifecycle steps, indicating the ongoing need for structured oversight and continuous metadata enhancement. This ensures that metadata remains relevant, accurate, and aligned with business needs. Furthermore, the mapping illustrates the increasing complexity and specificity of fine-grained features as the lifecycle progresses. For example, in the planning stage, the focus is on foundational aspects, such as rules and classification, while later

stages, including use and maintenance, introduce more technical elements, such as data lineage, task automation, and quality assessments. Features such as sensitive data detection and data quality assessments highlight the importance of compliance and accuracy in managing metadata.

7. RQ4: Mapping data catalog features to big data architectures requirements

Big data architectures have evolved towards metadata-driven architectures. Data lakehouse (Zaharia et al., 2021), data lake (Jahnke and Otto, 2023a), and data mesh (Dehghani and Fowler, 2022) have emerged as prominent paradigms. We selected these architectures as they have unique requirements concerning metadata management. Table 9 summarizes those requirements and maps them to the features of data catalogs. The following subsections provide a brief description of each big-data architecture and its potential use of data catalog features.

7.1. Data lake architecture

A data lake architecture is a data storage system designed to handle large volumes of structured, semi-structured, and unstructured data (Jahnke and Otto, 2023a). Unlike traditional data warehouses,

Table 8
Mapping of high-level and fine-grained features to metadata lifecycle management steps.

Lifecycle step	High-level features	Fine-grained features	Sources
Plan	Governance, Metadata Augmentation	Rules and Policies, Roles and Responsibilities, Data Risk, Business Glossary, Sensitive Data Detection, and Classification.	Anon (2025), Government of Canada (2024), Yang (2023), Bratt et al. (2017)
Construct, Create, Acquire	Metadata Harvesting, Metadata Augmentation, Metadata Usage.	Data Source Connectivity, Data Dictionary, Classification, Documentation, Business Glossary, and Storage.	Potvin and Thompson (2016), Anon (2025), Government of Canada (2024), Yang (2023), Bratt et al. (2017)
Commission, Organize, Store	Metadata Augmentation, Governance, Metadata Usage.	Documentation, Classification, Data Lineage, Documentation, Data Product, Business Glossary, and Storage.	Potvin and Thompson (2016), Anon (2025), Government of Canada (2024), Eichler et al. (2022), Yang (2023), Bratt et al. (2017), Stansberry et al. (2020)
Access	Metadata Usage, Collaboration.	Search, Access Controls, Sharing, Texting, and Data Product.	Potvin and Thompson (2016), Anon (2025), Government of Canada (2024), Eichler et al. (2022), Yang (2023), Bratt et al. (2017), Stansberry et al. (2020)
Use	Data Assessment, Collaboration, Metadata Harvesting.	Data Sampling, Data Dictionary, Data Quality Information, Rating, Reviewing, Dashboards, and Data Lineage.	Potvin and Thompson (2016), Anon (2025), Government of Canada (2024), Eichler et al. (2022), Yang (2023), Bratt et al. (2017), Stansberry et al. (2020)
Assess	Metadata Augmentation, Governance.	Data Quality Assessment, Rules and Policies, Sensitive Data Detection, and Change Tracking.	Potvin and Thompson (2016), Anon (2025), Government of Canada (2024), Bratt et al. (2017)
Maintain	Data Governance, Metadata Harvesting, Collaboration.	Task Automation and Standardization, Data Dictionary, Data Lineage, Data Quality Information and Assessment, Reviewing, Change Tracking, and Storage.	Potvin and Thompson (2016), Anon (2025), Government of Canada (2024), Yang (2023), Bratt et al. (2017), Stansberry et al. (2020)
Retire	Governance, Metadata Augmentation.	Data Risk, Classification, Rules and Policies, Changes Tracking, and Documentation.	Potvin and Thompson (2016), Anon (2025), Government of Canada (2024), Bratt et al. (2017)

which enforce strict schemas, data lakes store raw data sets in their native format, allowing for diverse data types such as logs, images, videos, and sensor data. It typically consists of multiple layers, starting with an ingestion layer that captures data from various sources, including databases, network devices, and streaming services. The storage layer leverages distributed storage systems, such as Hadoop HDFS,¹⁸ AWS S3,¹⁹ or Azure Data Lake Storage,²⁰ to retain both raw and processed data. A processing layer enables data transformation and analytics using frameworks for large-scale data processing, such as Apache Spark²¹ and Hadoop.²² Finally, the consumption layer provides access to data through business intelligence tools, machine learning frameworks, and query engines such as Presto²³ or Apache Hive.²⁴ Effective metadata management and governance are crucial for data lakes to maintain data quality and prevent becoming data swamps, ensuring that data remains discoverable, secure, and usable for analytical applications (Sawadogo and Darmont, 2021b).

7.2. Data lakehouse architecture

Data Lakehouse architecture is a hybrid data architecture that integrates the functionalities and benefits of data lake and data warehouse architectures, supporting structured and unstructured data in a unified platform (Zaharia et al., 2021). Similar to the data lake architecture, the data lakehouse architecture uses low-cost storage (e.g., Hadoop file

system or cloud storage buckets) and open file formats (e.g., Apache Parquet) for cost-effective and flexible data management (Mazumdar et al., 2023; Tagliabue et al., 2023). Compared with the data lakes, this architecture supports ACID (Atomicity, Consistency, Isolation, and Durability) database transactions, guaranteeing data integrity and reliability (Tagliabue et al., 2023). Data lakehouse also supports schema evolution, allowing changes to a data database schema without impacting the existing data while maintaining the compatibility of the old schema (Mazumdar et al., 2023). It utilizes metadata to define logical tables over physical files and uses these tables to enforce ACID transaction support and schema evaluation. One crucial component of the lakehouse architecture is a data catalog (called metastore) to store table names, schemas, and references to each table's specific metadata.

7.3. Data mesh architecture

Dehghani and Fowler (2022) defines data mesh as:

“a decentralized sociotechnical approach to share, access, and manage analytical data in complex and large-scale environments within or across organizations”

Data mesh decentralizes data ownership, assigning it to individual business domains to enhance flexibility and scalability. It treats data as a product, with each domain responsible for the quality, reliability, and usability of its data, including defining data access interfaces and ensuring data meets user needs. Data Mesh employs a federated governance model, striking a balance between central policies and domain-specific requirements to achieve tailored governance strategies. This architecture is organized around business domains, aligning data management practices with business processes and objectives (Dehghani and Fowler, 2022; Goedegebuure et al., 2024).

¹⁸ https://hadoop.apache.org/docs/r1.2.1/hdfs_design.html

¹⁹ <https://aws.amazon.com/it/s3/>

²⁰ <https://learn.microsoft.com/en-us/azure/storage/blobs/data-lake-storage-introduction>

²¹ <https://spark.apache.org/>

²² <https://hadoop.apache.org/>

²³ <https://prestodb.io/>

²⁴ <https://hive.apache.org/>

Table 9

Big data architectures requirements: ✓supported, ~ partially supported, xnot supported.

Requirement	Data Catalog Features	Lakehouse	Lake	Mesh	Ref.
Unified Data Platform	Data Source Connectivity, Data Dictionary, Data Inventory, Documentation, Data Lineage, Dashboards.	✓	✓	×	Mazumdar et al. (2023), Zaharia et al. (2021), Dehghani and Fowler (2022), Jahnke and Otto (2023a)
High Concurrency and Low Query Latency	Storage.	✓	~	~	Dehghani and Fowler (2022), Mazumdar et al. (2023)
Data Ingestion and Storage Layers	Data Source Connectivity, Data Dictionary, Data Sampling, Data Quality Information.	✓	✓	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Dremio (2021)
ACID Transactions	Storage.	✓	~	~	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023)
Data Governance and Security	Roles and Responsibilities, Rules and Policies, Task Automation and Standardization, Data Risk, Classification, Sensitive Data Detection, Data Lineage.	✓	×	✓	Dehghani and Fowler (2022), Jahnke and Otto (2023a), Dremio (2021), Mazumdar et al. (2023), Tagliabue et al. (2023)
Self-Service Data Access	Data Collaboration, Classification, Search, Data Dictionary.	✓	×	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Jahnke and Otto (2023a)
Schema Evolution	Documentation, Changes Tracking, Data Dictionary, Data Source Connectivity, Storage.	✓	~	×	Dehghani and Fowler (2022)
Cost Optimization and Scalability	Data Source Connectivity, Storage.	✓	✓	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Jahnke and Otto (2023a)
Unified Data Access	Data Source Connectivity and Data Dictionary.	✓	×	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Jahnke and Otto (2023a)
Data Virtualization	Data Sampling and Data Source Connectivity.	~	×	~	Dremio (2021)
Metadata Management	Documentation, Data Dictionary, Data Lineage, and Classification.	✓	✓	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Dremio (2021)
Real-Time Data Integration	Data Source Connectivity, Storage.	×	~	×	Dremio (2021)
AI and Machine Learning Integration	Task Automation and Standardization, and Dashboards.	✓	✓	×	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023)
Data Quality Management	Data Quality Assessment, Data Quality Information, Data Dictionary, Data Risk, Classification, and Data Lineage.	✓	~	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Dremio (2021)
Decentralized Data Ownership	Roles and Responsibilities, Data Product, and Data Dictionary.	×	×	✓	Dehghani and Fowler (2022)
Data as a Product	Data Product, Data Dictionary, Documentation, and Classification.	~	×	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023)
Federated Data Governance	Roles and Responsibilities, Rules, and Policies.	×	×	✓	Dehghani and Fowler (2022)
Domain-Oriented Data Architecture	Business Glossary, Data Dictionary, and Classification.	×	×	✓	Dehghani and Fowler (2022)
Interoperability and Standardization	Task Automation and Standardization, Tagging, Storage, and Classification.	✓	×	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Jahnke and Otto (2023a)
Scalable Data Infrastructure	Data Source Connectivity, Storage, and Dashboards.	✓	×	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Jahnke and Otto (2023a)
Automated Data Pipelines	Task Automation and Standardization.	✓	✓	✓	Dehghani and Fowler (2022), Mazumdar et al. (2023), Tagliabue et al. (2023), Jahnke and Otto (2023a)

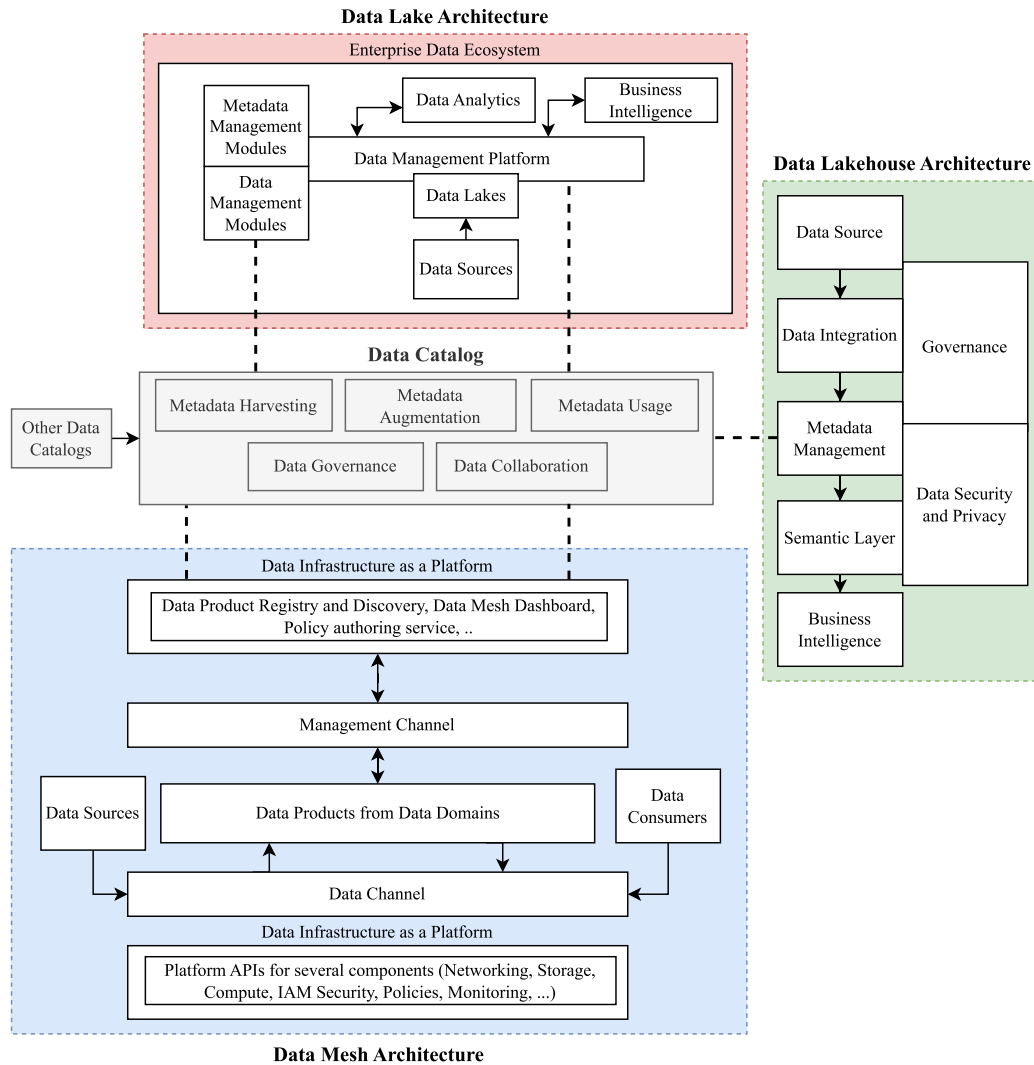


Fig. 11. Data catalog positioning within big data architectures: data lake (Jahnke and Otto, 2023a), data lakehouse (Harby and Zulkernine, 2022), data mesh (Goedegebuure et al., 2024). It also depicts the possible integration of a data catalog with other data catalogs.

7.4. Data catalog positioning within big data architectures

A data catalog is a key component in modern data architectures for metadata management, discovery, and, partly, data governance, which are essential capabilities for organizations handling large-scale data across data lakes, data lakehouses, and data mesh environments. Fig. 11 visualizes the positioning of the data catalog within these big data architectures.

In data lakes, where raw and semi-structured data is ingested from multiple sources, a catalog harvests metadata, enriches them with business context, and ensures discoverability, preventing the system from turning into a data swamp. It classifies and structures datasets, enabling analysts and data engineers to locate relevant data easily while enforcing access control, data lineage, and compliance policies (Jahnke and Otto, 2023a).

In data lakehouses, a data catalog enhances functionalities beyond basic metadata storage. A global e-commerce company, for instance, can utilize the Data Source Connectivity feature in a catalog to access real-time inventory data from warehouses worldwide, ensuring consistent and reliable reporting across regions. The catalog also plays a role in automated governance, defining roles and responsibilities for different teams, enforcing data integrity constraints, and enabling versioning

and auditing of transactional changes. Additionally, it facilitates self-service access, as seen in marketing teams that use the collaboration, search, and data dictionary features of the catalog to retrieve campaign performance data without requiring IT intervention.

In data mesh architectures, a data catalog serves as a centralized registry for distributed data products, ensuring discoverability, interoperability, and compliance with governance across domains. Each business unit can maintain its own glossary, classifications, and policies while still adhering to enterprise-wide governance frameworks. Features such as automated metadata standardization, policy-driven access controls, and lineage tracking enable teams to govern their data products autonomously while maintaining consistency throughout the organization.

The diagram (Fig. 11) further highlights how a data catalog integrates with data sources, governance frameworks, analytics tools, and security policies, creating a structured data management ecosystem across data architectures.

8. Discussion

This section discusses the implications of our findings for practitioners and outlines an attempt to roadmap potential future work from a research and innovation perspective.

8.1. Implications for practitioners

Our findings can help organizations systematically choose a data catalog tool based on their big data architectures and metadata management use cases. They also help tool developers identify features that enable them to gain competitive advantages.

8.1.1. Selecting the right data catalog tool

We identified 24 fine-grained and 5 high-level features that can be used to compare and assess the maturity of data catalog tools (see Sections 4, 5, and 4.6). Out of the 39 tools we analyzed, only one provides all the features we identified, while only five contain more than 90% of the fine-grained features. Considering the vast number of data catalog tools available on the market, our findings enable practitioners to make informed decisions when choosing a particular tool for their metadata use cases.

The tools we studied vary in their capabilities and maturity. Most tools offer the core capabilities necessary to build a data inventory, including data source connectivity, dataset search, dataset documentation, data lineage, classification, and detection of sensitive data. However, limited tools support key data governance features (e.g., data quality assessment and data dictionary) and collaboration features (e.g., commenting, reviewing, and texting). Moreover, the maturity of a given feature can also vary, for example, the number and type of data sources supported in the data source connectivity feature and natural language-based search queries in the search feature. In this respect, practitioners' work in harnessing our results should probably start from feasibility studies and sufficiently limited proof-of-concept pilots that determine baseline values for the above.

8.1.2. Integrating data catalogs into organizational big-data architectures

Our findings include the mappings between data architectures and catalog features (see Section 7). They can help organizations that need to choose a cataloging tool to implement the metadata-related requirements of their data architectures. However, organizations must carefully consider the potential mismatches between the quality requirements expected by a particular data architecture and what a specific deployment of a cataloging tool can offer. For instance, the data mesh architecture, which revolves around data products, defines new organizational roles and responsibilities, shifting data ownership to the data producer. In such a scenario, an all-in-one data catalog, such as a data marketplace, is not necessarily the best solution, as it undermines decentralization, one of the pillars of data mesh (Dehghani and Fowler, 2022; Goedegebuure et al., 2024).

8.1.3. Guidelines for effectively managing organizational metadata

Our results demonstrate the complexity and interdependence of data catalog capabilities required for effective metadata lifecycle management within an organization (see Section 6). The distribution of high-level and fine-grained features at each lifecycle stage underscores the need for a holistic approach to metadata management, rather than relying on isolated tools. For example, the governance feature is present in nearly every stage, emphasizing that effective oversight, compliance, and risk management must be integrated throughout the entire lifecycle. The consistent appearance of the metadata augmentation as a feature across stages suggests a need for adaptive and evolving metadata, which are continuously enriched to meet changing organizational needs. We also observed that the collaboration features are notably concentrated in the access and use metadata lifecycle stages. However, the effectiveness of these features depends on their integration with other lifecycle stages. Automation and standardization also emerge as crucial capabilities that support the scalability and operational efficiency of metadata management within an organization. Catalog features, such as task automation, metadata standardization, and machine learning integration, can help organizations manage the growing volume and complexity of their data assets. These features reduce the manual effort required for metadata management, allowing catalogs to scale with the organization's data needs.

8.1.4. Tool feature prioritization

Our research enables data catalog tool developers to evaluate the maturity of their tools in terms of their support for the requirements of modern big data architectures and holistic metadata management processes within organizations (Section 4.6). For example, features such as real-time data integration, dynamic schema evolution, data product, and data quality assessment are underrepresented in the tools we analyzed. By closing these gaps, these tools can expand their capabilities to meet the diverse needs of various use cases within organizations.

8.2. Research challenges and the road ahead

Our results provide insights into the potential research directions related to data cataloging. We identified three key directions: collaborative and AI-driven capabilities of data catalogs, data governance in emerging big data architectures, and tool interoperability.

8.2.1. Advanced data catalog features

We observed that data catalogs employ AI to implement features such as contextual metadata enrichment, metadata search and recommendation, and predictive data governance. These features would make data catalogs more proactive, helping organizations anticipate data needs, recommend relevant datasets, and surface insights tailored to user context (Petrik et al., 2023).

Research on the collaboration features of data catalogs, such as texting, sharing, reviewing, and rating, can assess how they impact teamwork and data usage, and identify ways to enhance their effectiveness and integration. Insights into user behavior, preferences, and collaboration barriers could drive the development of more impactful, integrated collaborative functionalities within data catalog environments (Jahnke and Otto, 2023b; Labadie et al., 2020).

An important aspect that emerged but remains underexplored is the maturity of data catalogs. Literature offers limited guidance on assessing or defining it (Immuta, 2025; Corcho et al., 2024; Travkin, 2024). As part of our future work, we plan to investigate this concept further through interviews with practitioners and organizations actively using data catalogs, to identify practical indicators and stages of maturity.

8.2.2. Data governance in modern data architectures

With the rise of decentralized architectures, governance across distributed data domains presents new challenges, including inconsistent policies and standards, global data visibility, compliance enforcement and auditing, and operational overhead. The researchers could focus on federated governance models that ensure consistent data policies, central visibility, and compliance monitoring while respecting domain autonomy (Dehghani and Fowler, 2022).

8.2.3. Tool interoperability

As organizations increasingly adopt hybrid and multi-cloud strategies, there is a growing need for interoperability across data catalogs from different vendors or platforms. Research could address the development of standardized metadata protocols, APIs, and data exchange formats that enable seamless interoperability between catalogs (Jahnke and Otto, 2023b).

9. Threats to validity

During our study, we identified several threats to validity. We addressed or mitigated most of them to ensure the reliability and robustness of our findings. We divided these threats into three categories: internal, external, and construct validity (Wohlin et al., 2012).

Internal validity. This validity concerns the validity of the methods employed to select and analyze the literature sources and data catalog tools. A potential threat is the risk of missing relevant literature sources and tools due to deficiencies in the search process. As discussed in Section 3, we systematically formulated the search queries based on research questions. Moreover, we conducted searches on the digital libraries and search engines used by existing literature review studies. We applied inclusion/exclusion and quality assessment criteria to select the most suitable sources from peer-reviewed journals, conferences, and grey literature. Researchers' personal biases could lead to misinterpretation of data catalog features and influence the analysis. To mitigate this threat, we implemented a systematic coding process, with regular reviews among authors, to maintain consistency and objectivity. We developed a clear and consistent framework through iterative refinement. We included grey literature sources and official tools documentation to avoid oversampling positive findings on data catalogs, to incorporate various perspectives.

External validity. Threats to external validity can challenge the generalizability of our findings. A potential external threat is the evolution of data catalog tools and grey literature sources, which can impact the longevity of some of our findings. To partially mitigate this risk, we attempted to identify data catalog features that can cover the entire metadata management lifecycle, which has been standardized and adopted by both practitioners and researchers. Another potential threat is excluding literature sources and tools that provide new insights. As discussed above, we systematically searched for sources and tools to mitigate this issue.

Another potential threat to the validity of our findings is the limited exploration of the maturity of data catalog implementations. While the concept appears relevant and potentially impactful, the reviewed literature provides limited information or frameworks for assessing maturity levels across different organizational contexts. As a result, our current understanding of this dimension remains preliminary. In future work, we plan to address this gap through follow-up interviews with practitioners and adopters of data catalogs.

Construct validity. We provided explicit definitions for each term to avoid misunderstandings about key concepts that could impact the accuracy of our mappings. Incorrect mapping of features to metadata life-cycle management, maturity levels, and big data architecture requirements may lead to flawed insights. To mitigate this threat, the first two authors of the paper collaboratively created and refined mappings over multiple interactions. When disagreements occurred, the other authors were involved in resolving them. The subjective nature of qualitative data analysis can lead to inaccuracies. To address this issue, we employed triangulation by cross-referencing findings from multiple data sources to validate interpretations.

10. Conclusion

This study aimed to identify the features of modern data catalogs and their roles within metadata life-cycle management and metadata-driven big-data architectures, such as data mesh, data lake, and data lakehouse. With this intention, we conducted an MLR (Multivocal Literature Review), analyzing 86 academic and grey literature articles, as well as the documentation of catalog tools.

First, we employed the selected literature to develop a classification framework for data catalogs comprising five high-level features and 24 fine-grained ones. Next, based on those features, we identified three maturity levels for data catalogs. We analyzed and classified 39 tools using the identified features and maturity levels. Finally, we mapped the features to the metadata management life cycle steps and modern big data architecture requirements to understand the potential usage of data catalogs in metadata management and data architectures.

Our results indicate that data catalogs have progressed beyond basic data indexing and search capabilities, incorporating essential functions

for data governance, quality assessment, and collaboration to meet the complex demands of modern data architectures and metadata use cases. Despite their advancements, gaps remain, particularly in supporting user collaboration, data and metadata governance, tool interoperability, and utilizing AI to enhance data catalog features. These limitations present opportunities for future research to improve data catalog capabilities. Addressing these areas will be crucial as organizations continue to scale their data operations and seek more decentralized, agile data management solutions.

As data catalogs evolve, they will play an increasingly central role in enabling organizations to derive value from data by providing a unified, governed, and accessible metadata environment. This will empower data-driven decision-making and foster collaboration across diverse data landscapes.

CRedit authorship contribution statement

Marco Tonnarelli: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Indika Kumara:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration. **Stefan Driessen:** Writing – review & editing, Validation, Supervision. **Damian Andrew Tamburri:** Writing – review & editing, Validation, Supervision, Project administration, Methodology. **Willem-Jan van den Heuvel:** Writing – review & editing, Supervision, Project administration. **Patrick Oor:** Supervision, Project administration, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors utilized ChatGPT by OpenAI and Grammarly to enhance the grammar, structure, and clarity of the content provided. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Marco Tonnarelli reports financial support was provided by NXP Semiconductors NV. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work has been founded by NXP Semiconductors under the project named RAPTURE (TKI HTSM/23.0367) with Eindhoven University of Technology - Jheronimus Academy of Data Science.

Marco Tonnarelli (corresponding author) and Damian Andrew Tamburri are part of it, Stefan Driessen and Indika Kumara are NOT part of it. Willem-Jan van den Heuvel and Patrick Oor are also involved in the project.

Data availability

In the article there is a link pointing to a Google Drive folder with the data used.

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